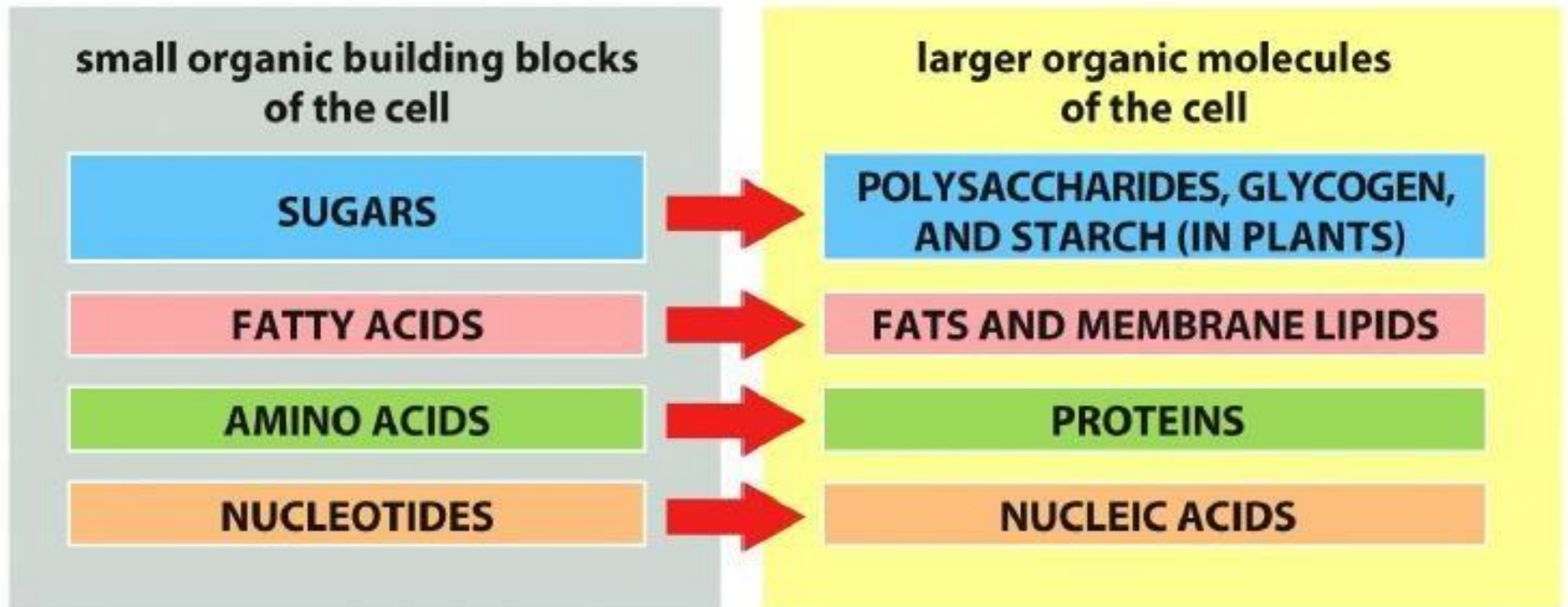
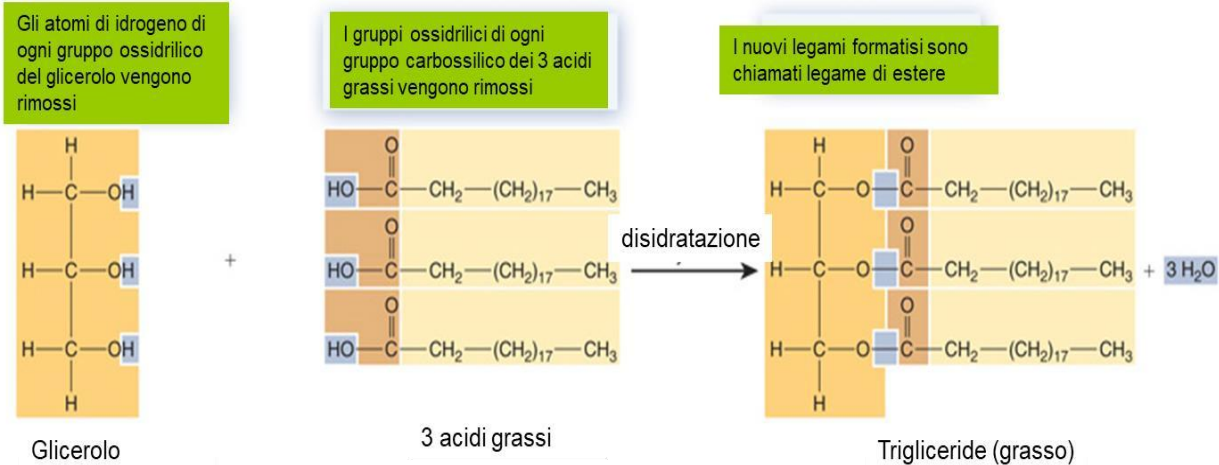
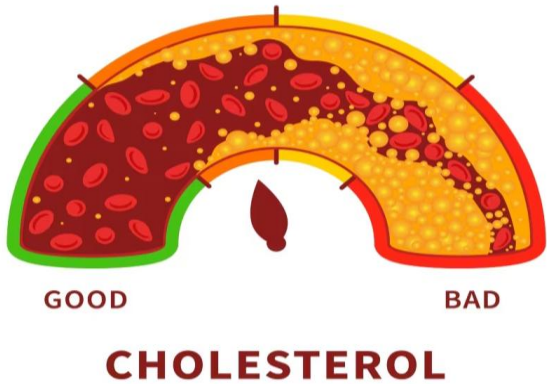
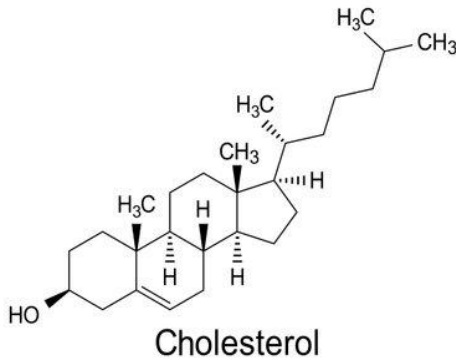


The macromolecules of the cell



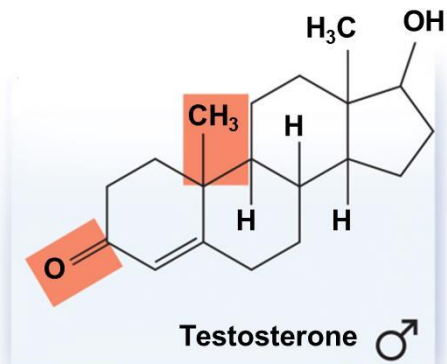
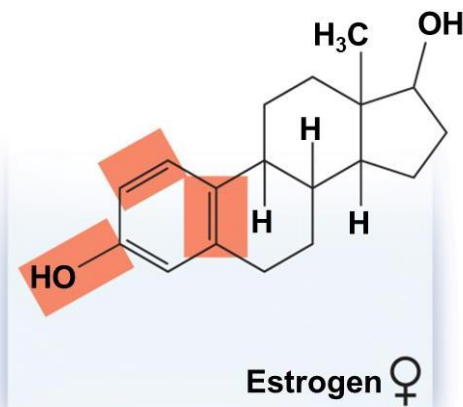
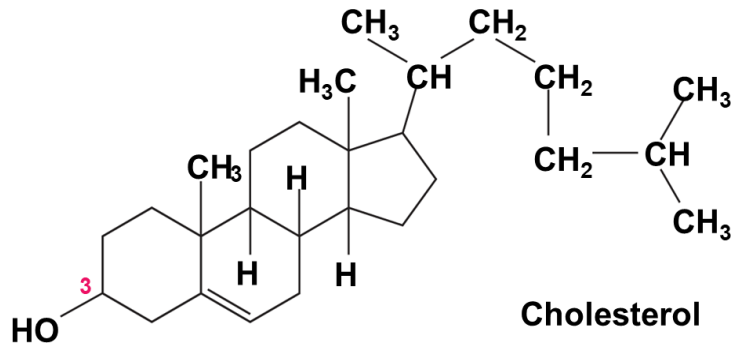
Lipidi - Grassi

- Composti principalmente da atomi di idrogeno (H) e carbonio (C)
- I lipidi sono non-polari e quindi insolubili in acqua



Lipidi - Grassi

Ormoni steroidei



Il **colesterolo** è la molecola da cui originano
gli ormoni sessuali:
Estrogeni e Androgeni



Female cardinal



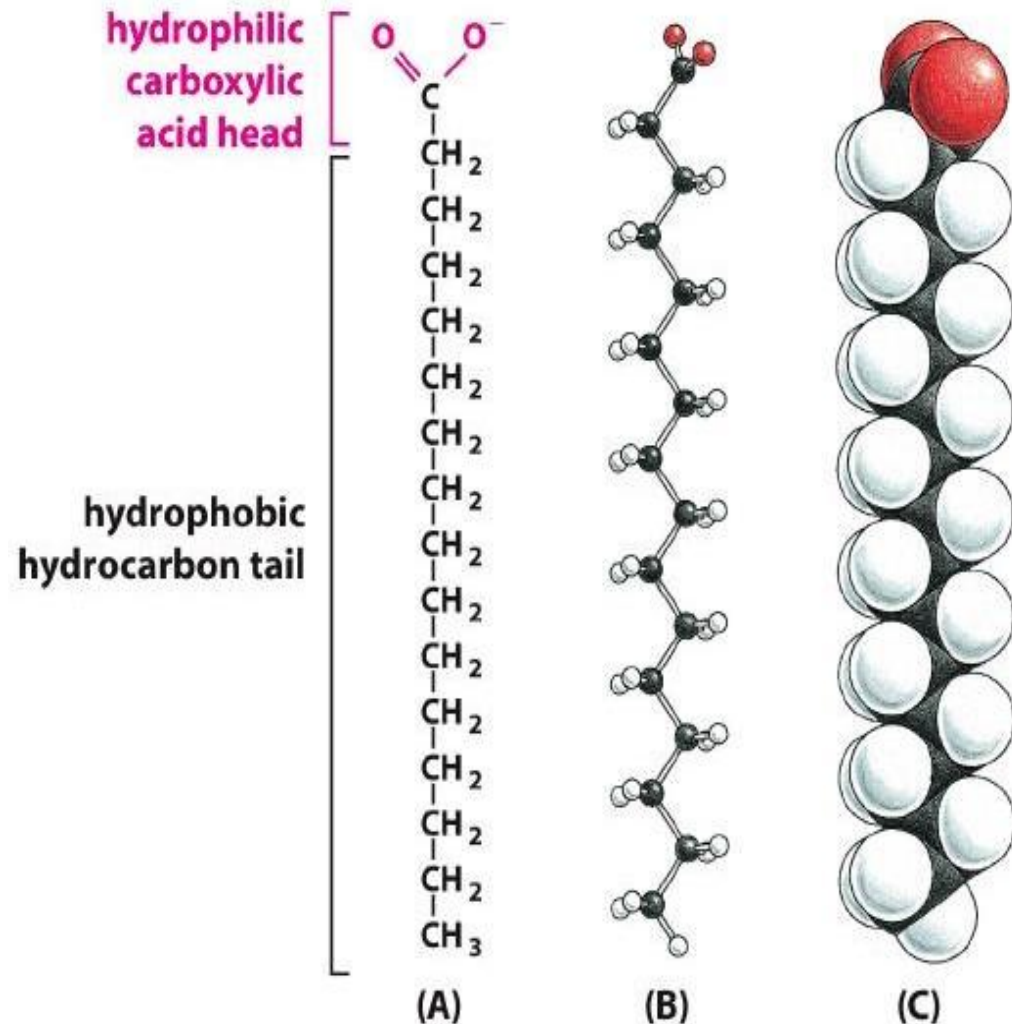
Male cardinal

Lipidi - Grassi

A parte infarto ed ictus.....

I lipidi sono fondamentali per:

1. La generazione della cellula stessa e dei vari organelli grazie al loro ruolo cruciale nella formazione delle membrane.
2. Immagazzinare energia (trigliceridi), 1 grammo di grassi immagazzina due volte più energia di 1 grammo di carboidrati.



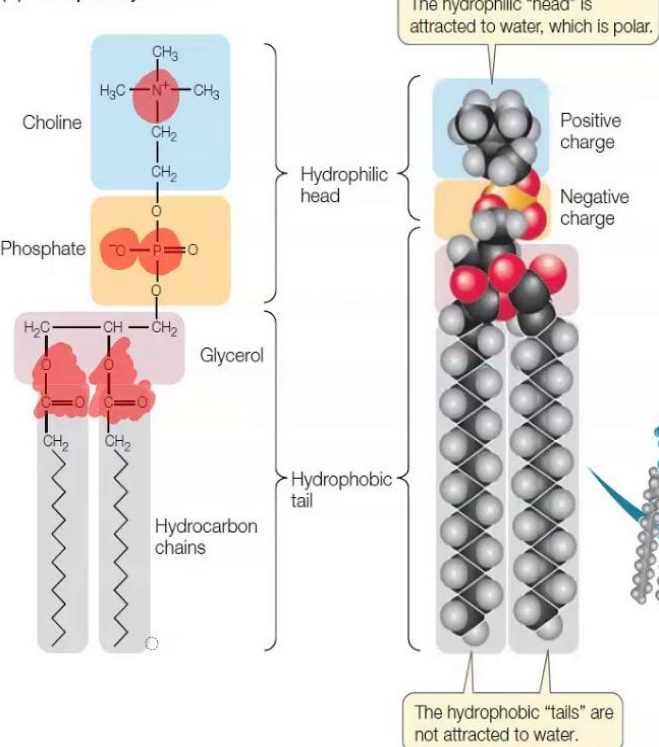
Lipidi - Grassi

FOSFOLIPIDI

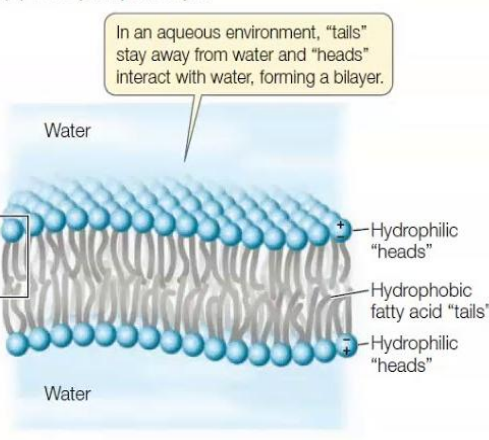
Struttura: 2 acidi grassi, glicerolo ed un gruppo fosfato al quale si legano specifiche molecole quali colina, inositolo, serina, etanolammina.

- 1. Regione del fosfato è polare ed idrofilica
- 2. Catene di acidi grassi sono apolari ed idrofobici

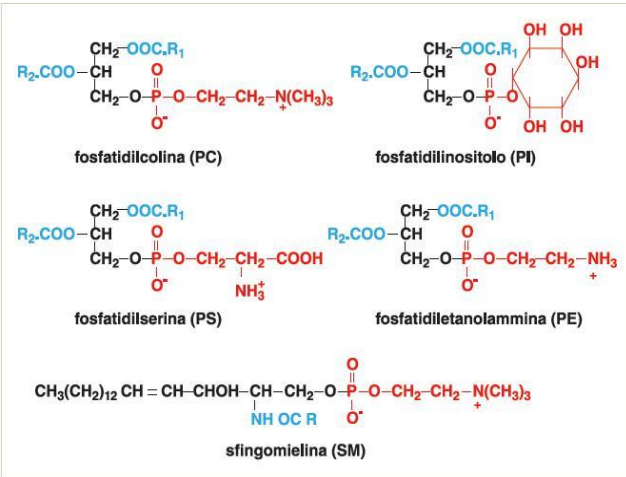
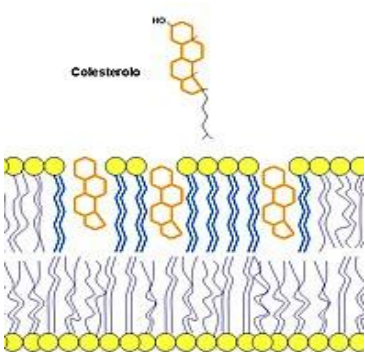
(A) Phosphatidylcholine



(B) Phospholipid bilayer



Il colesterolo aumenta l'ordine del tratto prossimale delle catene idrofobiche, diminuendo l'isomerizzazione trans-gauche e gli inginocchiamenti transitori delle catene stesse.



Carboidrati

I carboidrati, detti anche glucidi (dal greco "glucos" = dolce) sono sostanze formate da carbonio ed acqua e sono contenuti principalmente negli alimenti di origine vegetale.

La forma più semplice di carboidrati sono i monosaccaridi o zuccheri semplici come glucosio, fruttosio, ribosio $C_n(H_2O)_n$. Sono molecole polari e quindi facilmente solubili in acqua. In base alla loro struttura chimica si distinguono in Aldosi e Chetosi.

MONOSACCHARIDES

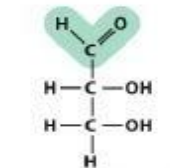
Monosaccharides usually have the general formula $(CH_2O)_n$, where n can be 3, 4, 5, or 6, and have two or more hydroxyl groups. They either contain an aldehyde group ($-C(=O)H$) and are called aldoses, or a ketone group ($>C=O$) and are called ketoses.

3-carbon (TRIOSES)

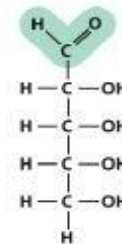
5-carbon (PENTOSES)

6-carbon (HEXOSES)

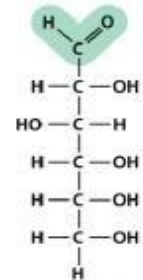
ALDOSES



glyceraldehyde

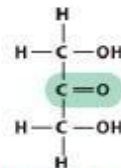


ribose

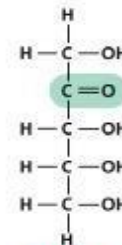


glucose

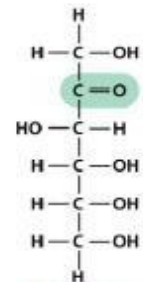
KETOSES



dihydroxyacetone



ribulose

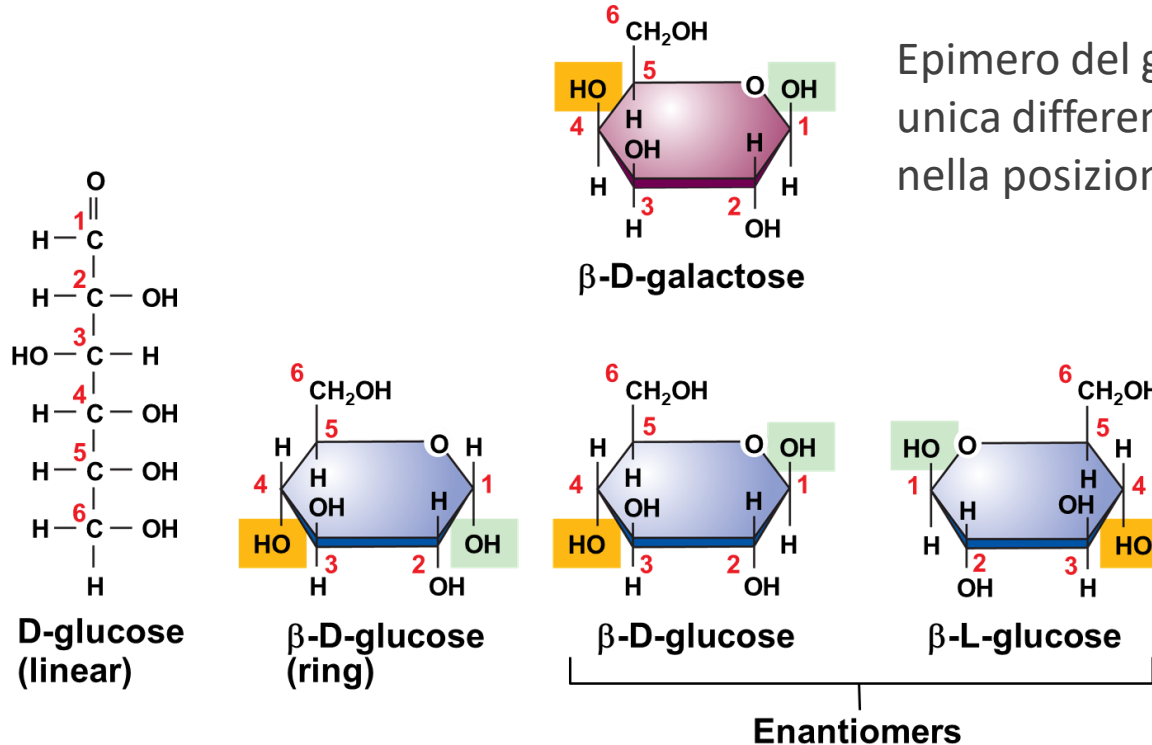


fructose

Carboidrati

RING FORMATION

In aqueous solution, the aldehyde or ketone group of a sugar molecule tends to react with a hydroxyl group of the same molecule, thereby closing the molecule into a ring.



Epimero del glucosio:
unica differenza fra i due esosi è
nella posizione dell'-OH sul C-4.

(a) Linear and ring structures
of α -D-glucose

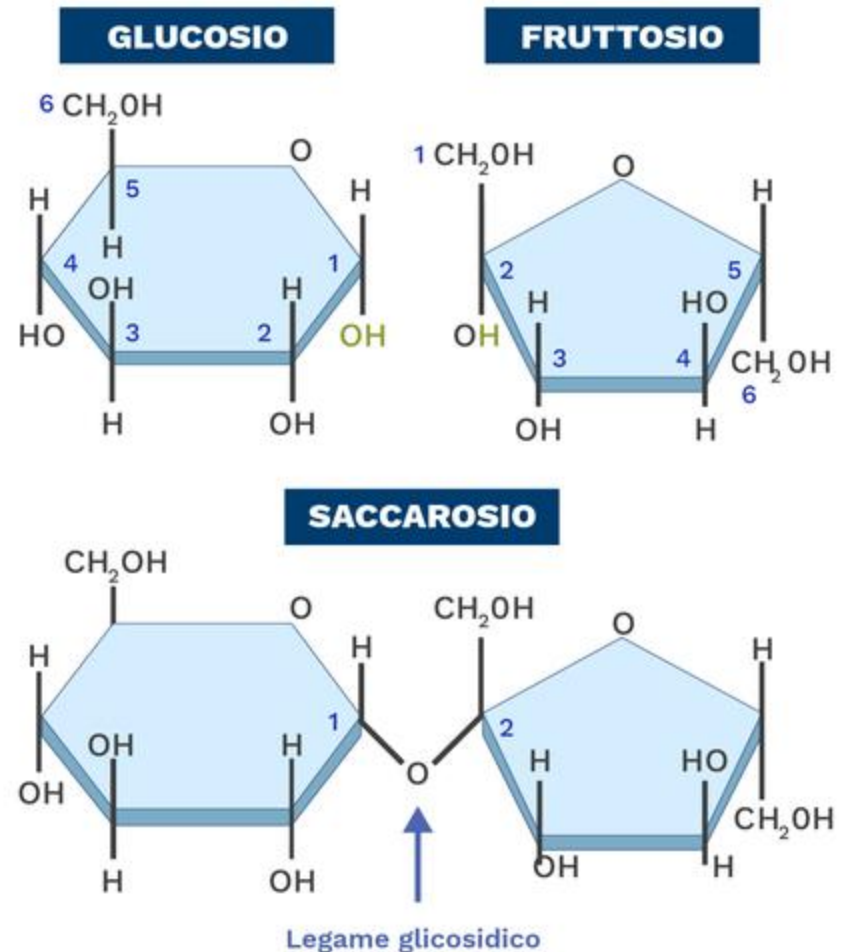
(b) Isomers of glucose

Carboidrati

Disaccaridi

Per formare un disaccaride due monosaccaridi si uniscono tramite un legame glicosidico tra il carbonio di un monosaccaride e il gruppo alcolico dell'altro monosaccaride con eliminazione di una molecola di acqua.

I **più comuni disaccaridi** sono: il **saccarosio**, zucchero da tavola (sintetizzato dalle piante attraverso il processo di fotosintesi clorofilliana) è costituito da una molecola di glucosio + una molecola di fruttosio; il **lattosio** (si trova nel latte di ogni mammifero) è costituito da una molecola di glucosio + una molecola di galattosio; il **maltosio**, costituito da due molecole di glucosio.

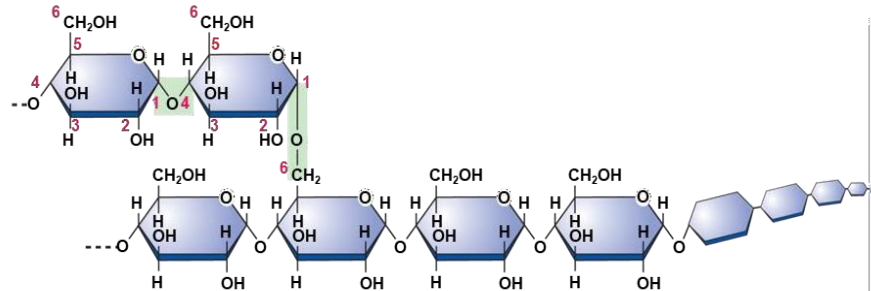


Carboidrati

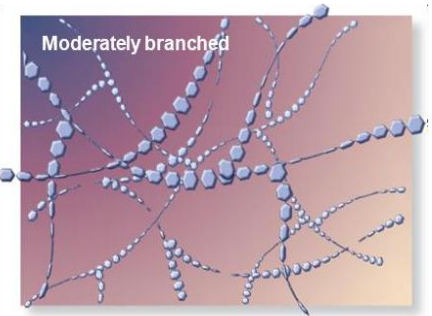
Polisaccaridi

Poco ramificato
poco solubile in H_2O

Starch [Amido]

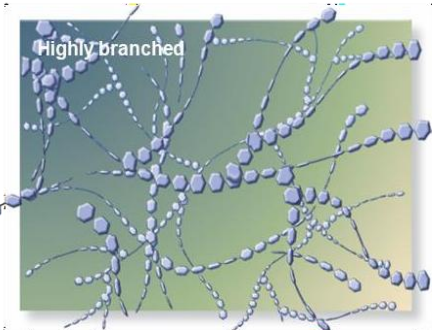
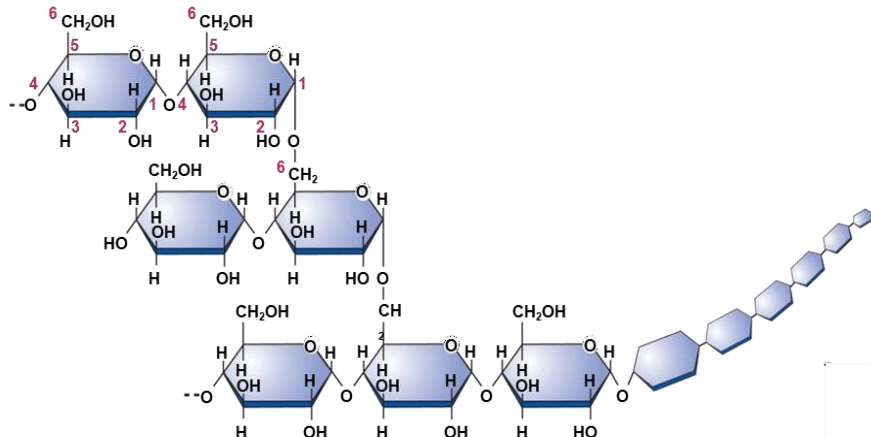


Branching patterns



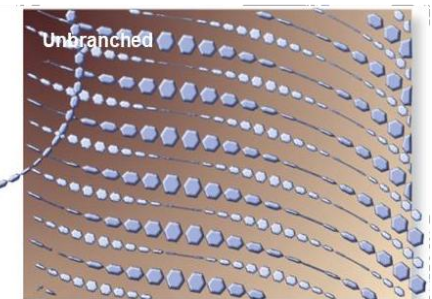
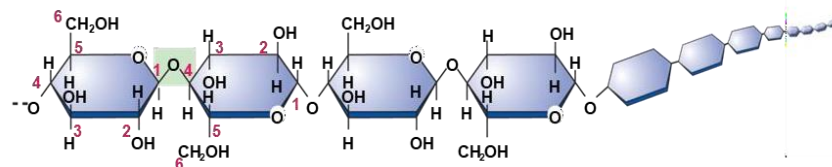
Molto ramificato
solubile in H_2O

Glycogen [Glicogeno]



Non ramificato
insolubile in H_2O

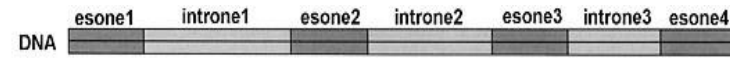
Cellulose [Cellulosa]

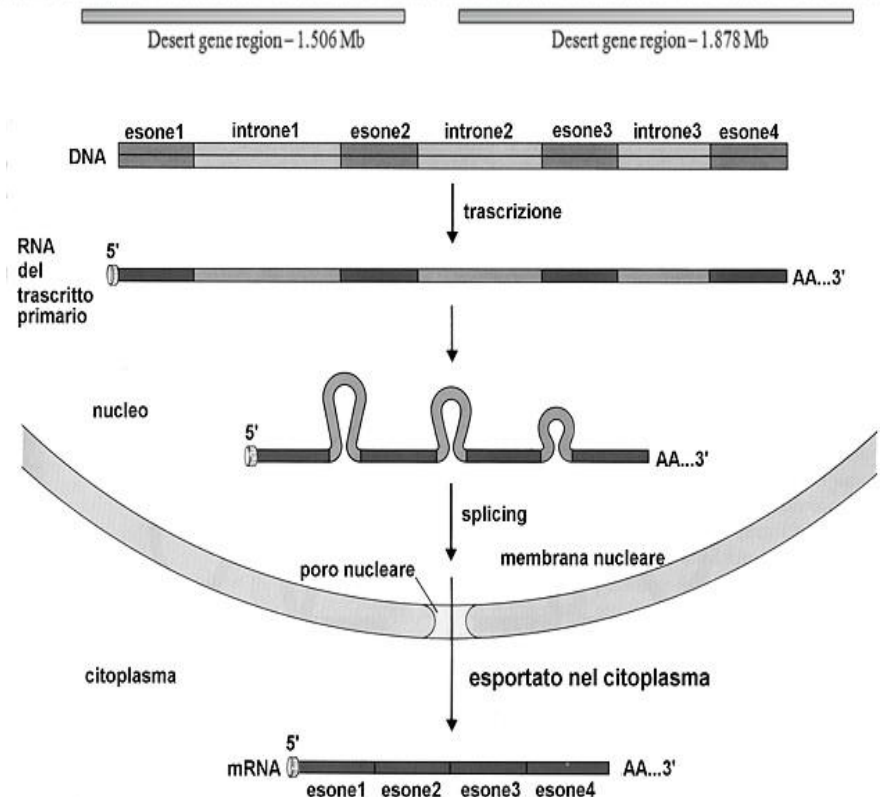
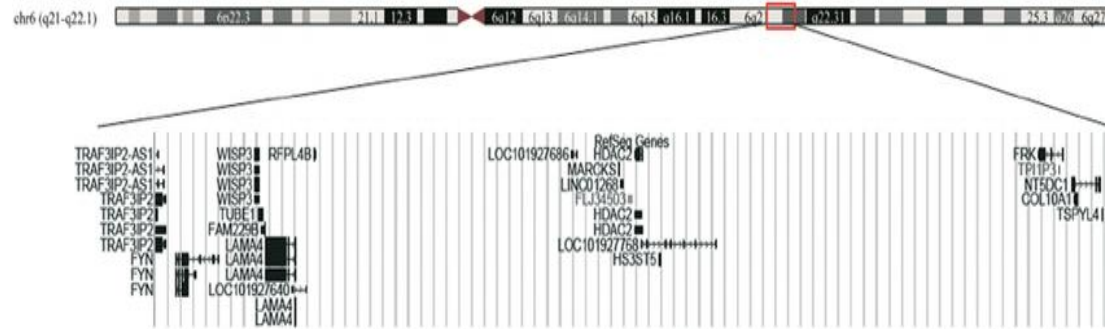


Acidi nucleici DNA & RNA

Responsabile dell'immagazzinamento, trasmissione ed utilizzo dell'informazione genetica.

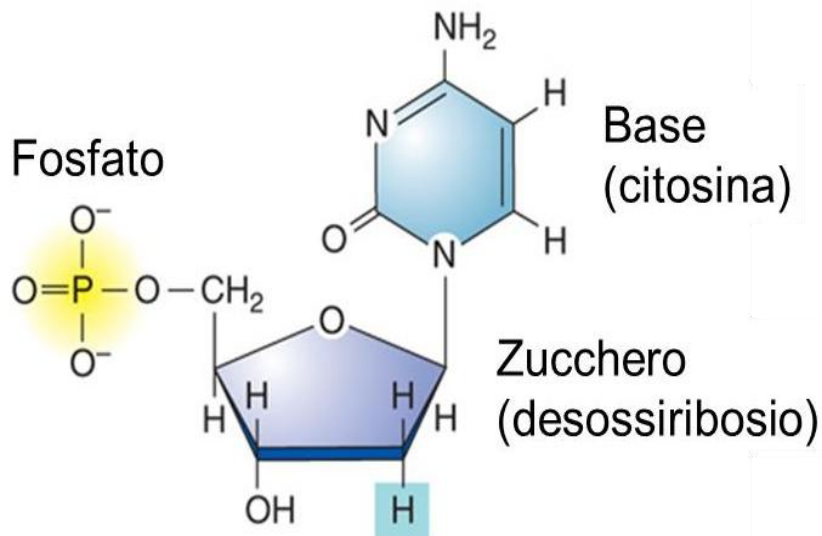
Se ne distinguono due classi:

- Acido deossiribonucleico (DNA)
 - Relegato all'interno del nucle immagazzina le informazioni genetiche codificate nella sequenza dei suoi monomeri
- Acido ribonucleico (RNA)
 - Coinvolto nella trasmissione di queste informazioni all'esterno del nucleo rendendole istruzioni funzionali per la cellula

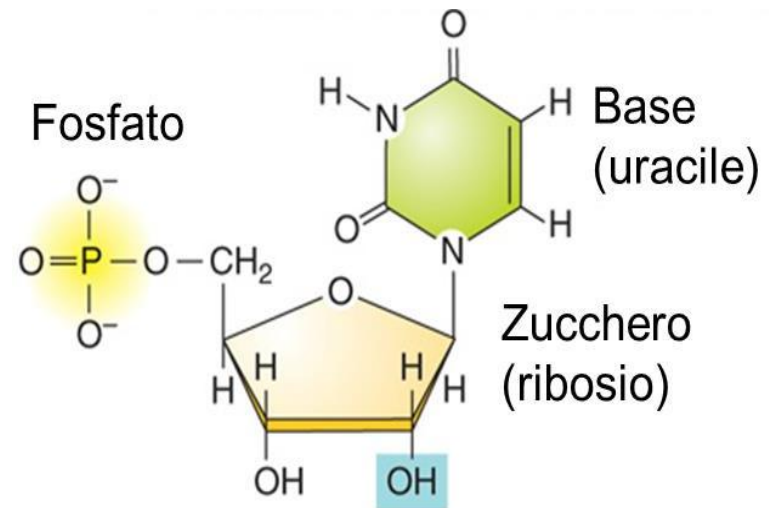


Acidi nucleici- I Nucleotidi

Il monomero degli acidi nucleici è il nucleotide che è costituito da:
uno zucchero a cinque atomi di carbonio (desossiribosio DNA e ribosio RNA), una base formata da un singolo o doppio anello contenente azoto (base azzotata), un gruppo fosfato.



Esempio di un desossiribonucleotide

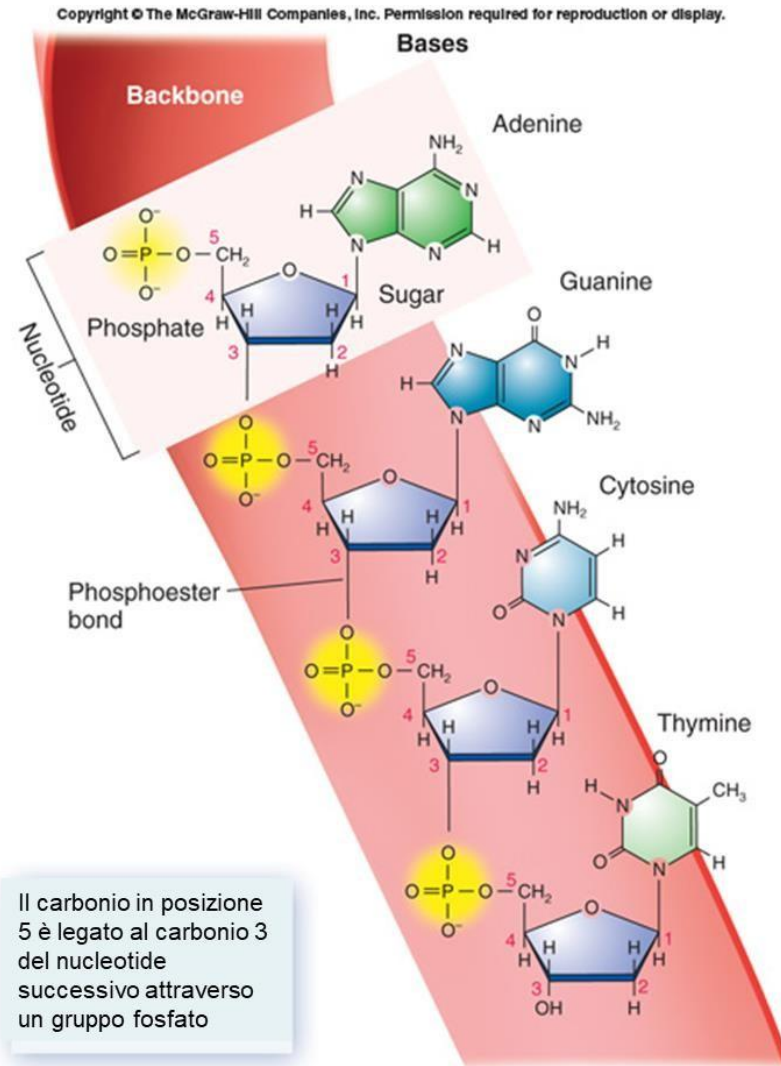
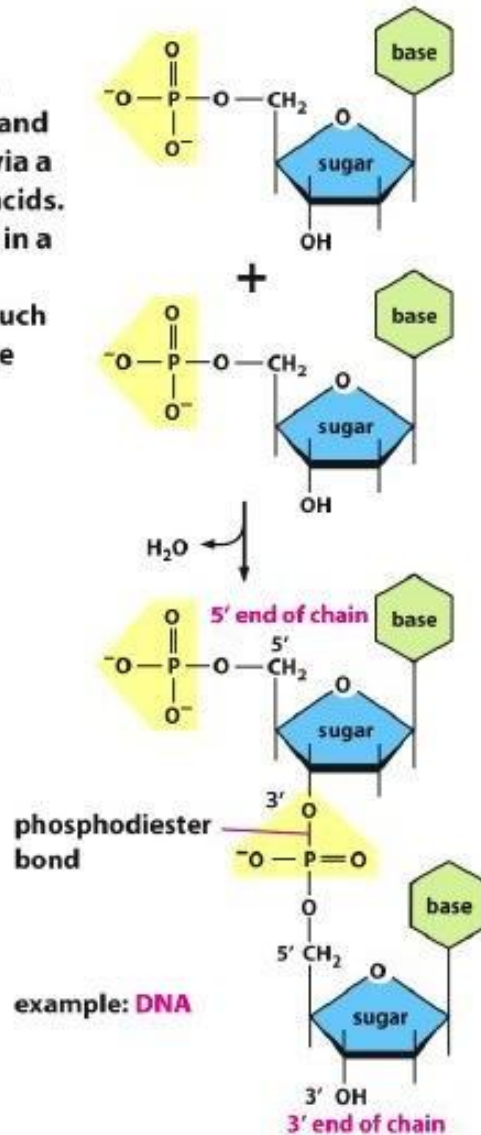


Esempio di un ribonucleotide

Acidi nucleici- Il Legame Fosfodiester

NUCLEIC ACIDS

Nucleotides are joined together by **phosphodiester bonds** between 5' and 3' carbon atoms of the sugar ring, via a phosphate group, to form nucleic acids. The linear sequence of nucleotides in a nucleic acid chain is commonly abbreviated by a one-letter code, such as AGCTTACA, with the 5' end of the chain at the left.



Acidi nucleici- Le Basi Azzotate

NOMENCLATURE

The names can be confusing, but the abbreviations are clear.

BASE	NUCLEOSIDE	ABBR.
adenine	adenosine	A
guanine	guanosine	G
cytosine	cytidine	C
uracil	uridine	U
thymine	thymidine	T

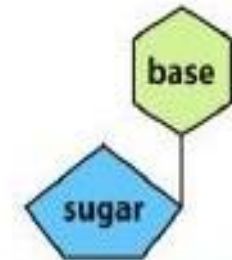
Nucleotides are abbreviated by three capital letters. Some examples follow:

AMP = adenosine monophosphate

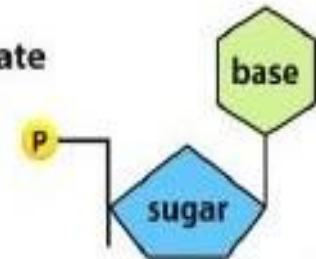
dAMP = deoxyadenosine monophosphate

UDP = uridine diphosphate

ATP = adenosine triphosphate

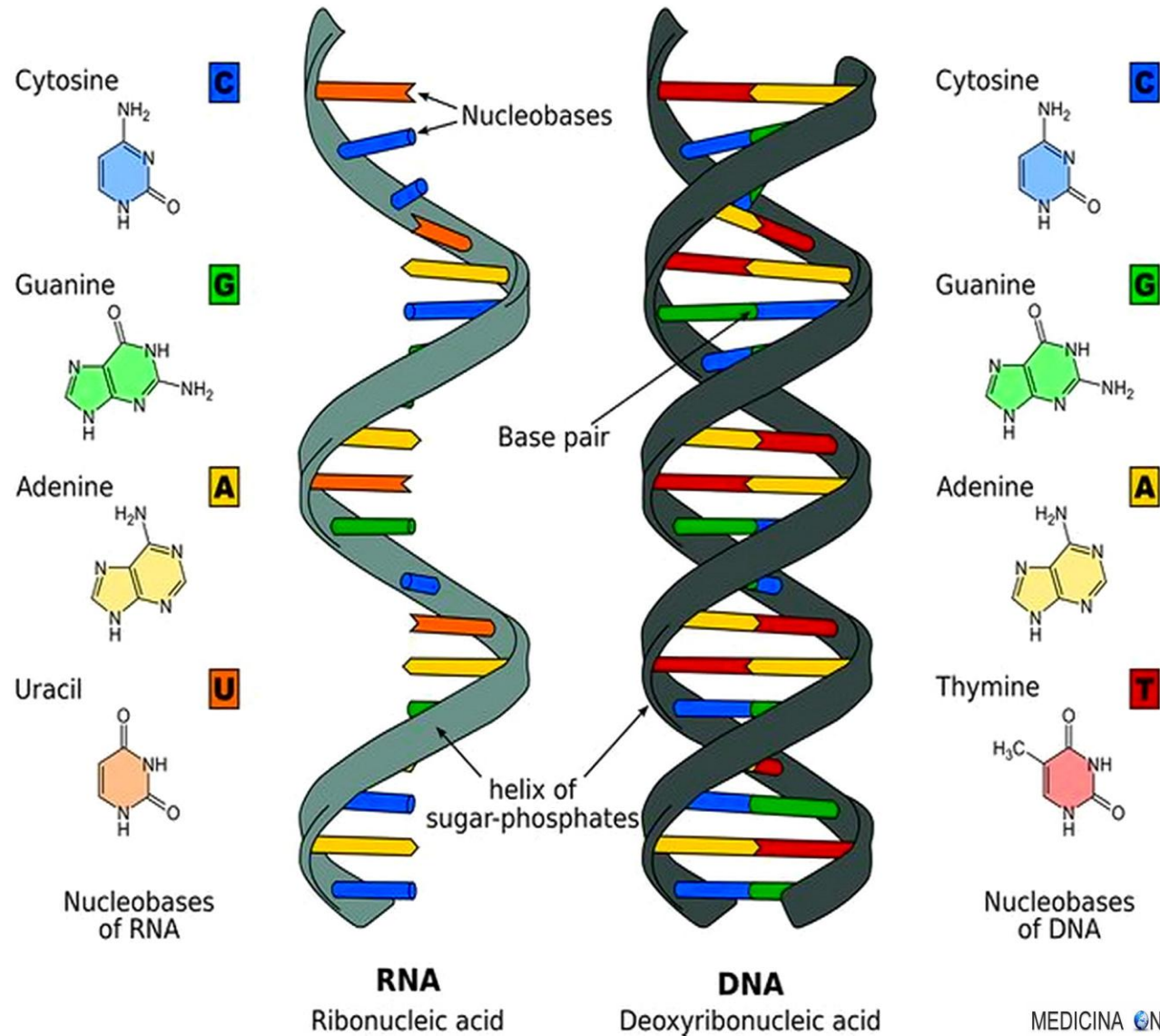


BASE + SUGAR = **NUCLEOSIDE**



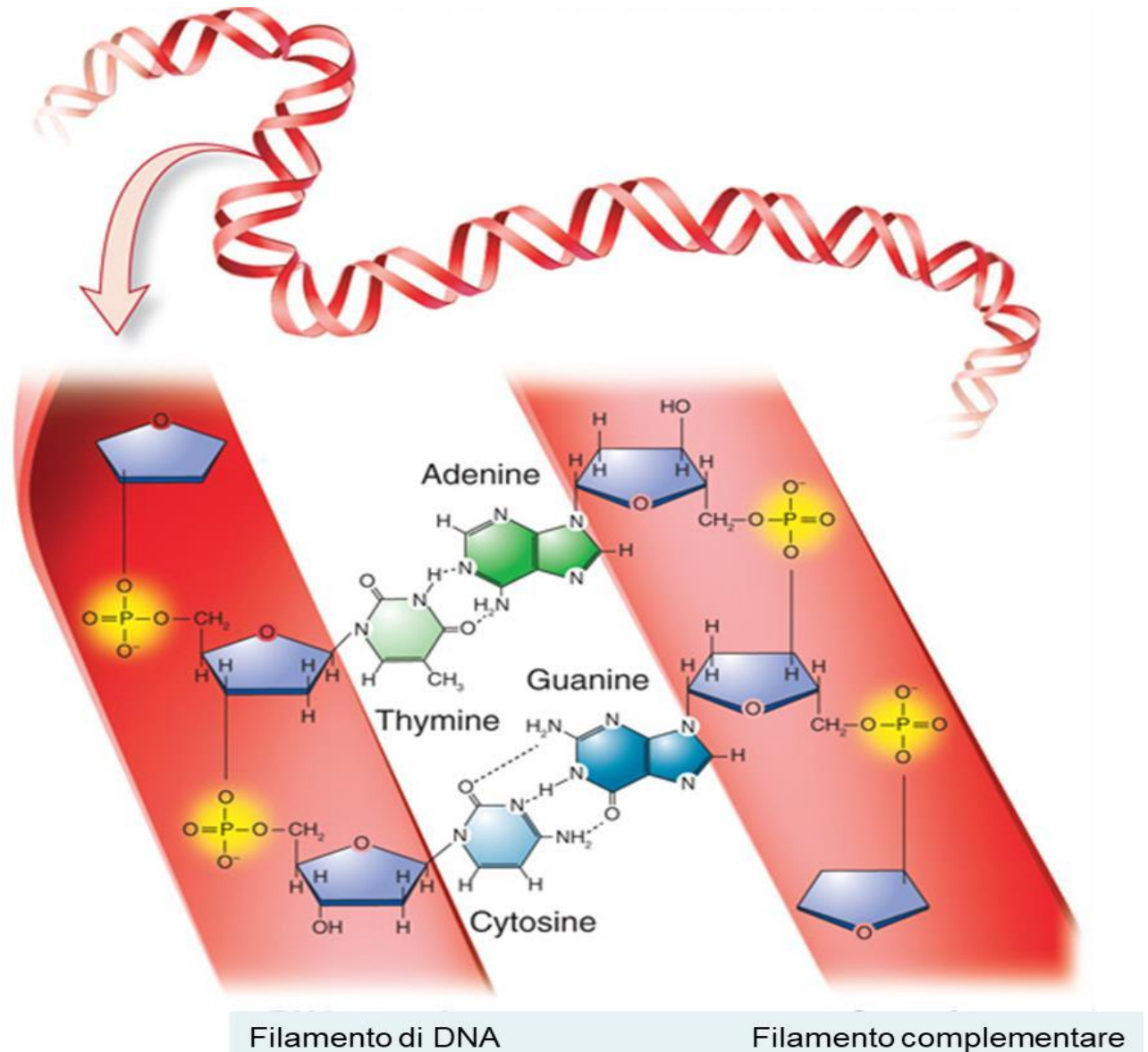
BASE + SUGAR + PHOSPHATE = **NUCLEOTIDE**

Acidi nucleici- Struttura-Interazioni-Funzioni



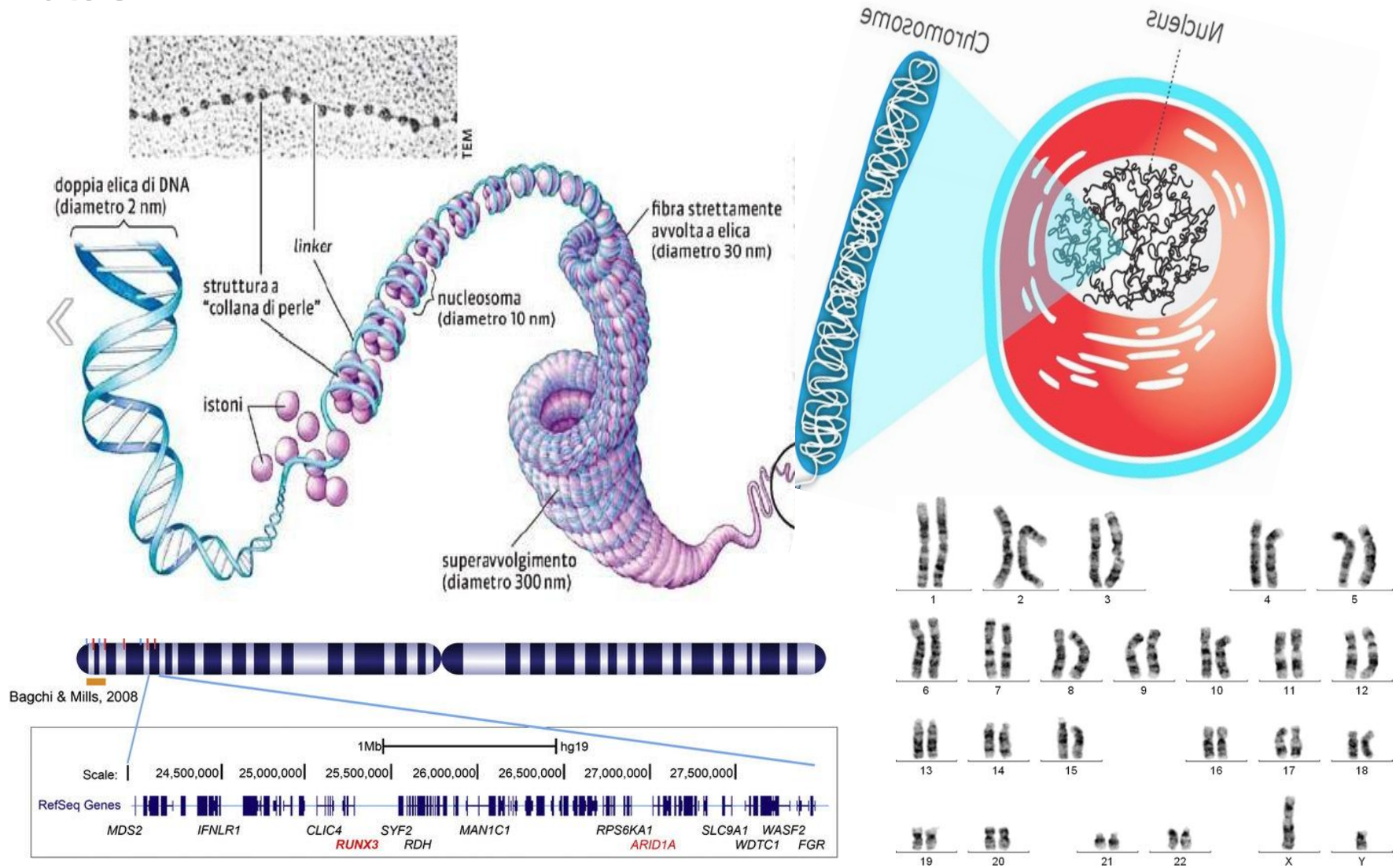
Acidi nucleici- Struttura-Interazioni-Funzioni

DNA



Acidi nucleici- Struttura-Interazioni-Funzioni

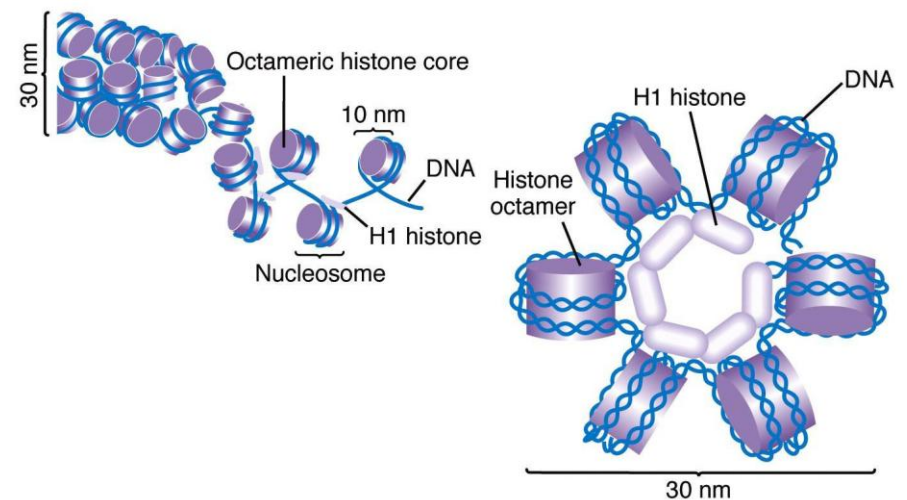
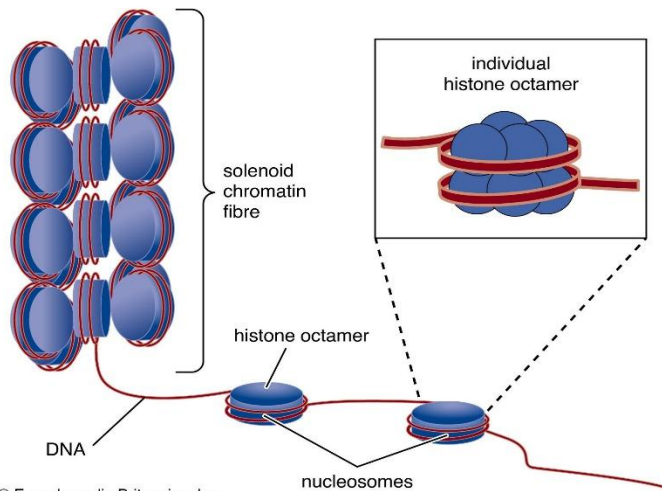
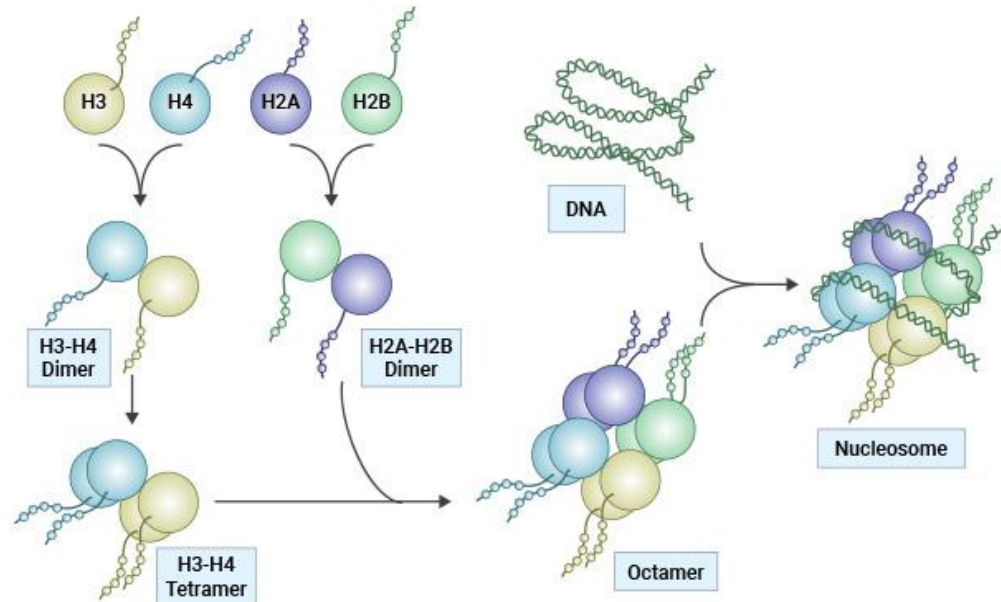
DNA



Acidi nucleici- Struttura-Interazioni-Funzioni

Nucleosome

- Simplest packing str of DNA
- 146 bp DNA wrapped around histone octamer
- Octamer = 2 copies of 4 core histones
- DNA length varies b/w species
- Core DNA – DNA associated with histone octamer
- Linker DNA – DNA b/w histone octamer – 8 to 114 bp



Acidi nucleici- Struttura-Interazioni-Funzioni

DNA

EUCROMATINA
ATTIVA

ETEROCROMATINA
INATTIVA

Porzioni di cromatina poco spiralizzata
Poco elettrendensa

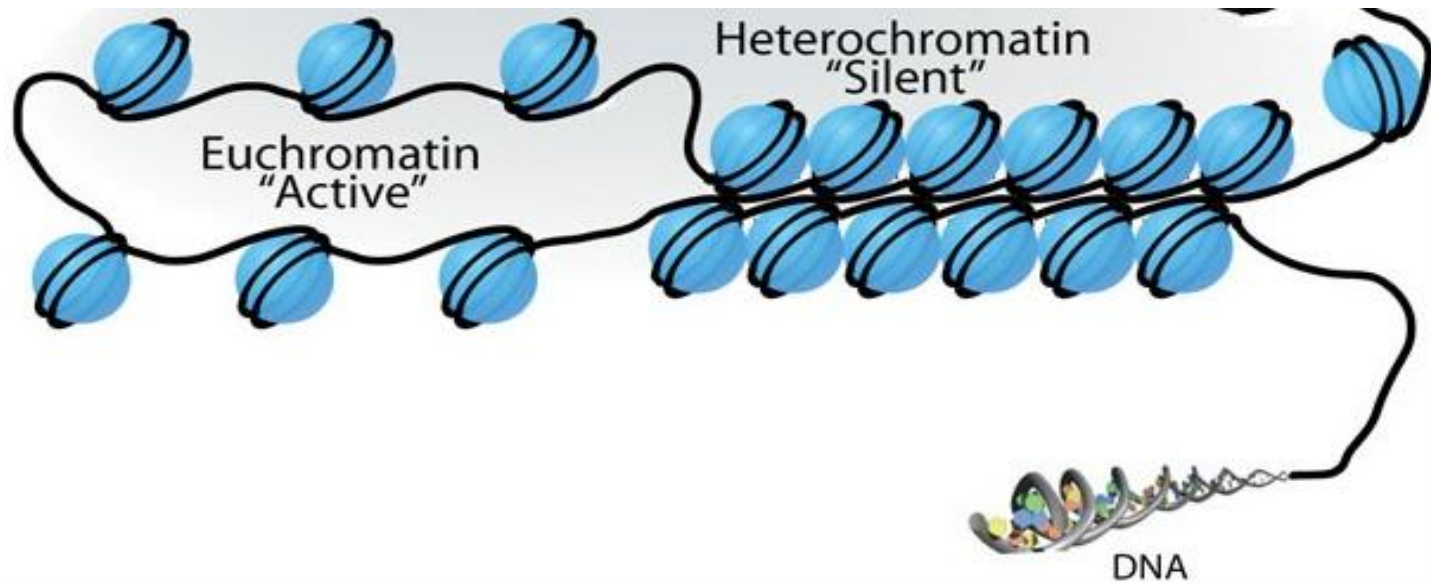


Il DNA è accessibile per il
macchinario della trascrizione

Porzioni di cromatina molto spiralizzata
Elettrendensa



Il DNA non è accessibile per il
macchinario della trascrizione

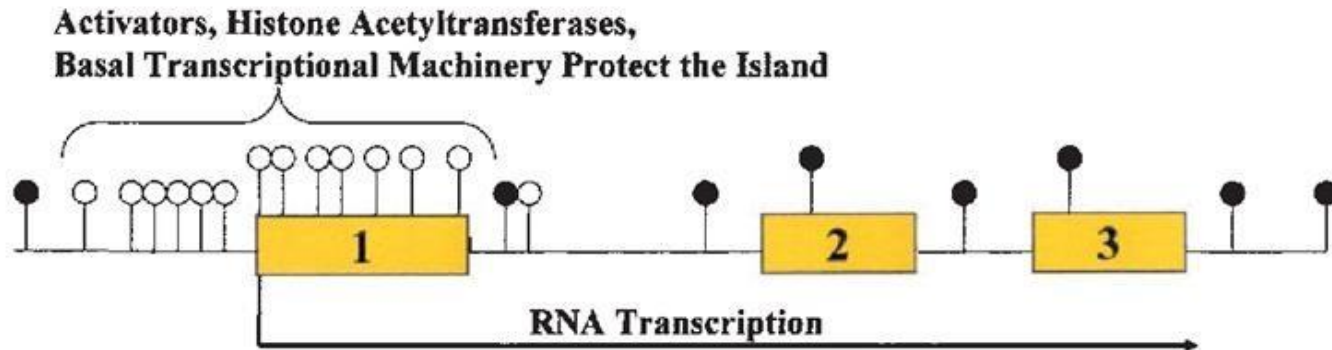


Modifiche del DNA

Responsabili di cambiamenti conformazionali della cromatina

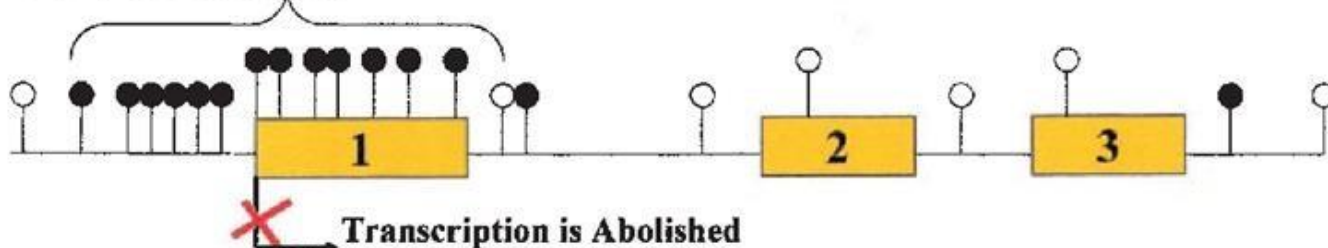
Dei circa 3 miliardi di coppie di basi che costituiscono il genoma di mammifero, **circa il 40% sono coppie CG e il 2-7% di esse è metilato**. Nelle cellule eucariotiche la metilazione è a carico della Citosina (C) delle isole CpG.

Unmethylated CpG Island



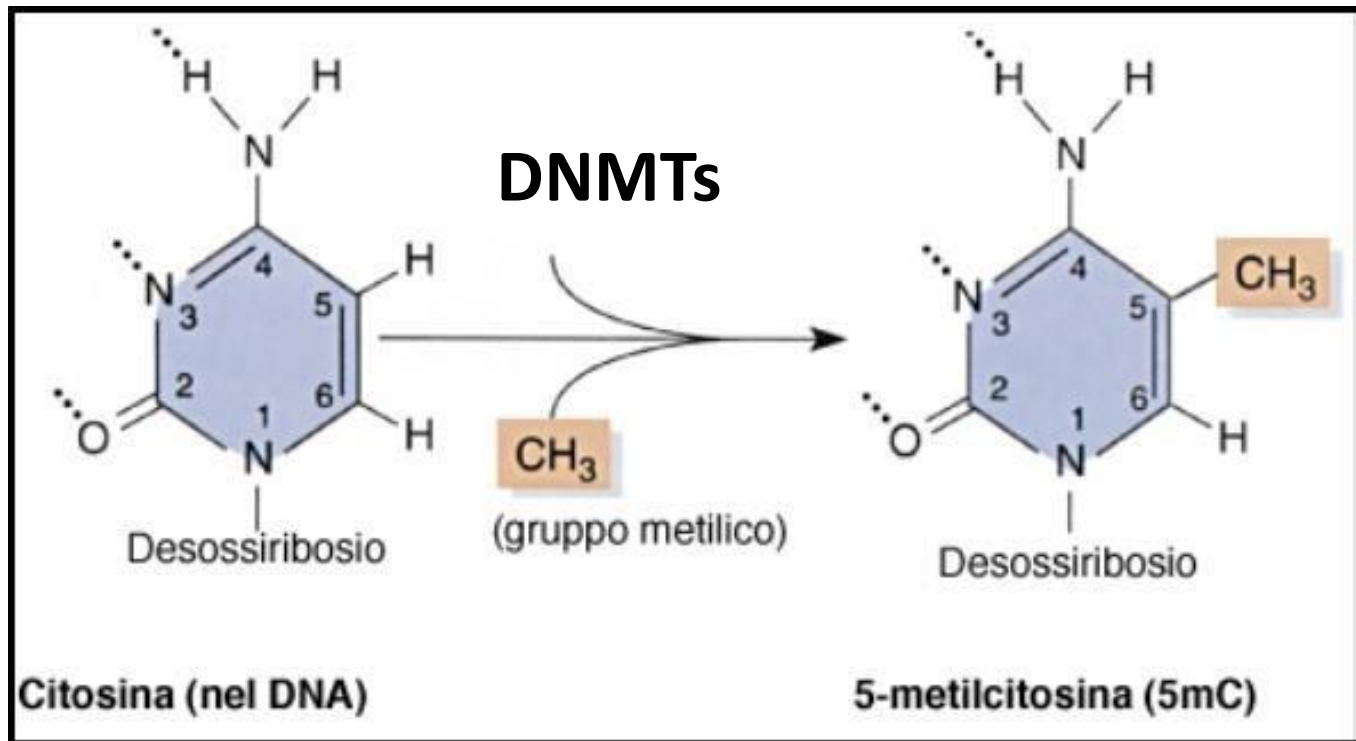
Hypermethylated CpG Island

Transcriptional Repressors, Histone Deacetylases,
DNA Methyltransferases and Methyl-binding Proteins
Shut-Down the Island



L'enzima DNA metiltrasferasi (DNMT) opera una metilazione in posizione 5 della base citosina, con formazione di 5-metilcitosina.

Anche queste sono famiglie di enzimi che presentano diverse isoforme (DNMT1, DNMT3a, DNMT3b..)

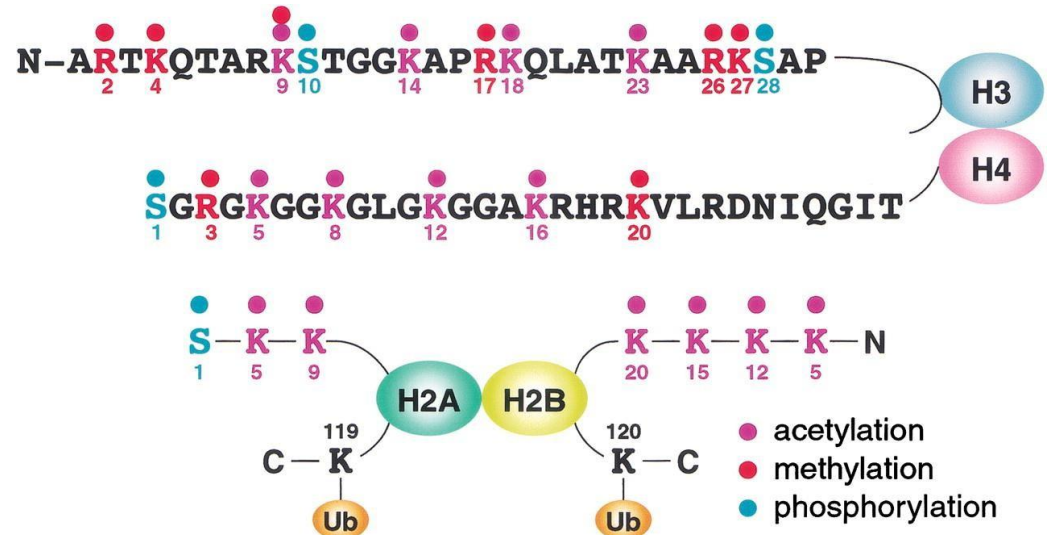
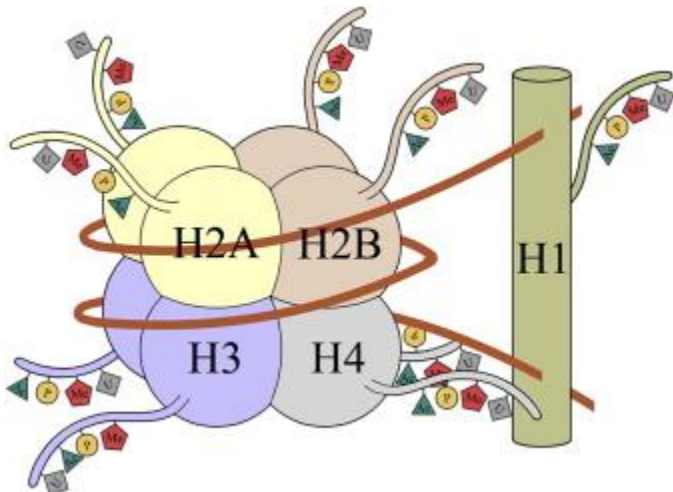
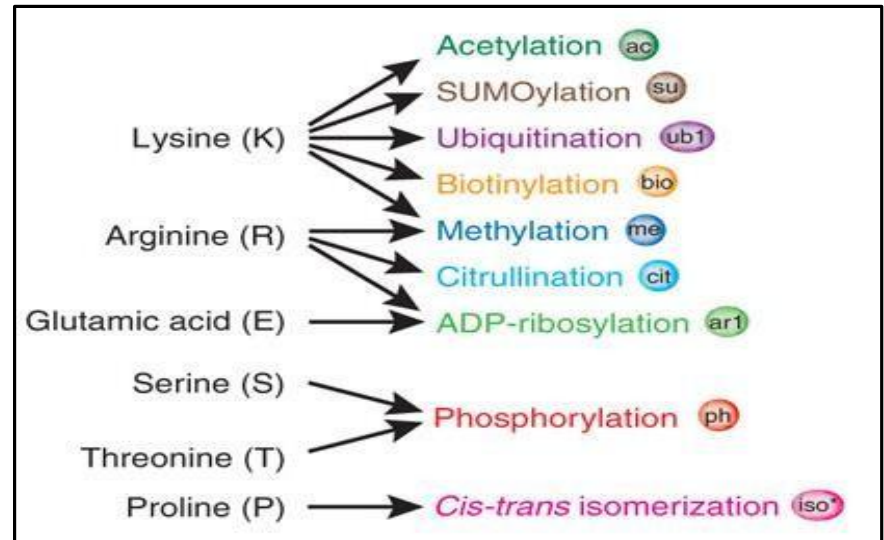


Isole CpG: tratti di genoma, nei quali le sequenze CpG sono 10 volte più frequenti, si trovano soprattutto in prossimità dei siti di inizio della trascrizione.

Modifiche degli istoni

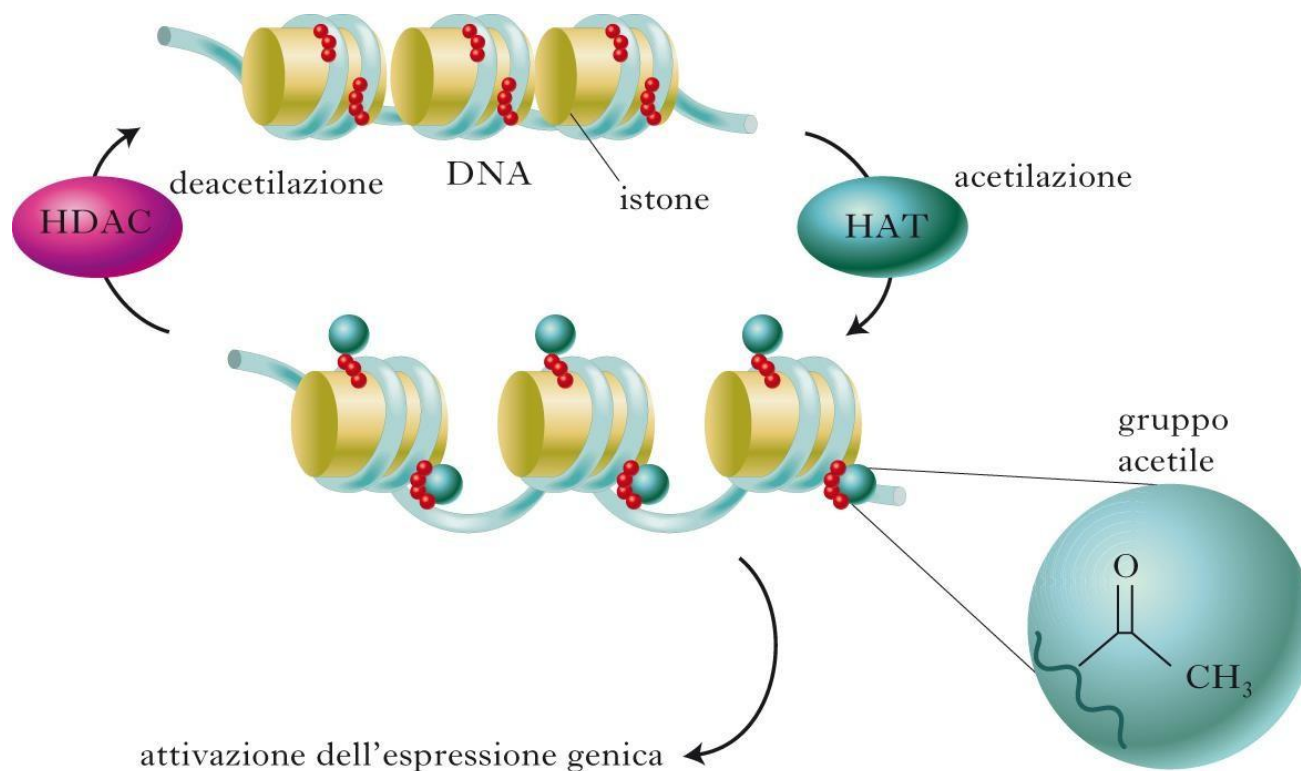
Responsabili di cambiamenti conformazionali della cromatina

- Acetilazione
- Metilazione
- Fosforilazione
- Ubiquitinazione
- Sumoilazione
- ADP-ribosilazione



Lisine = K; Arginine = R; Serine = S

Acetilazione e De-Acetilazione



HDACs

La deacetilazione degli istoni è associata al condensamento della cromatina ed alla conseguente

→ Inibizione della trascrizione genica

HATs

L'acetilazione degli istoni è associata al decondensamento della cromatina ed alla

→ Trascrizione genica

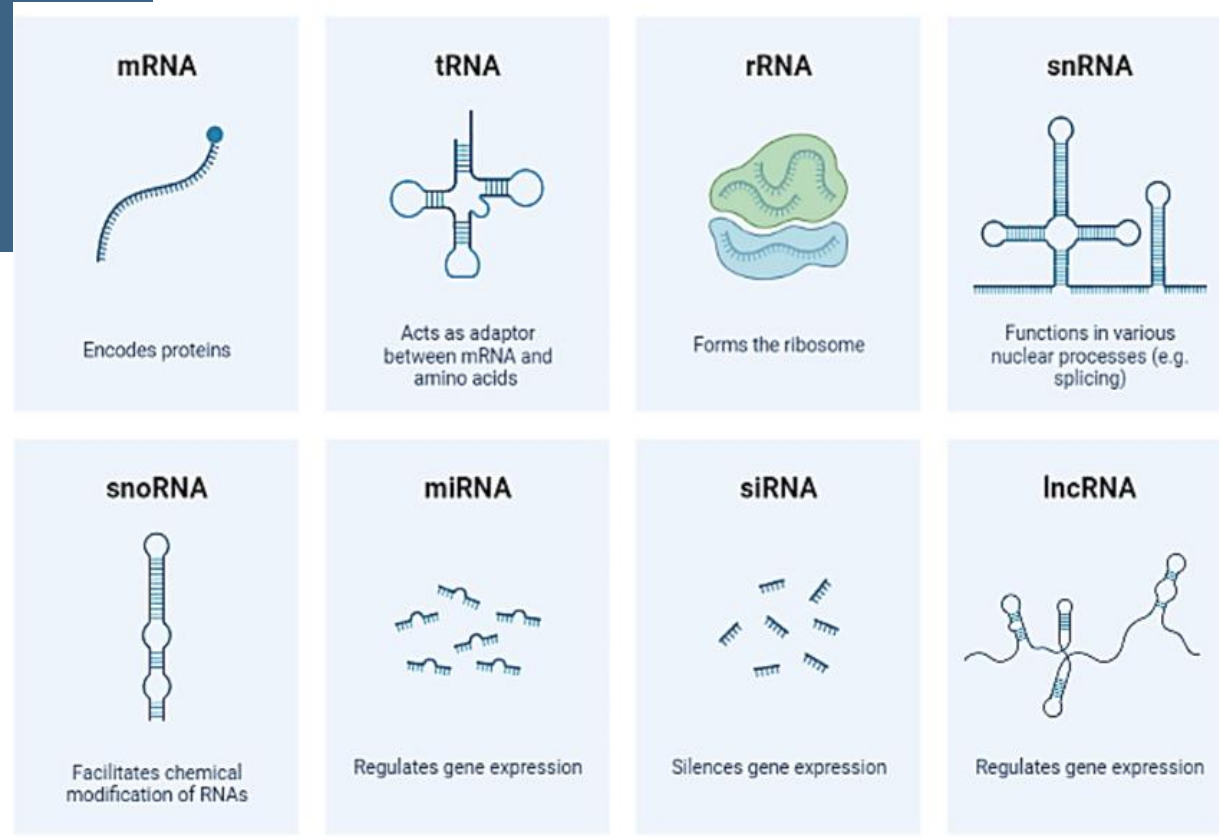
Acidi nucleici- Struttura-Interazioni-Funzioni

RNA



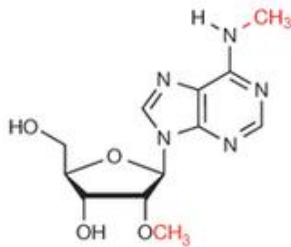
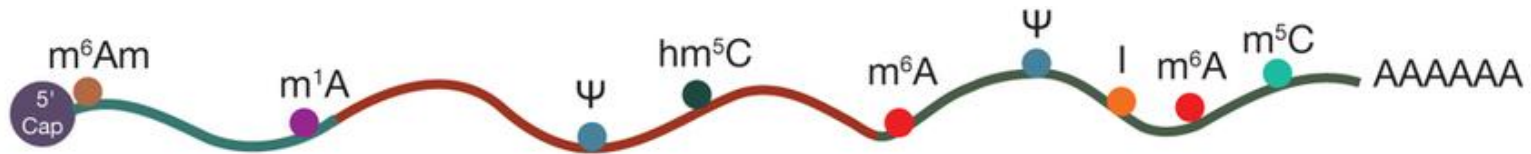
Strutture secondary e terziarie

Molteplici funzioni

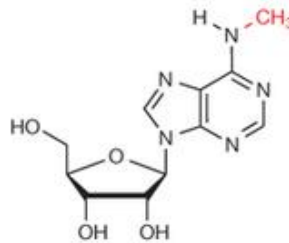


Modifiche del RNA

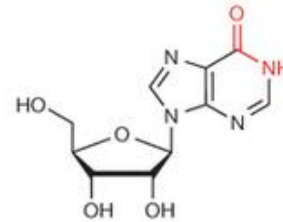
Responsabili di cambiamenti funzionali della molecola



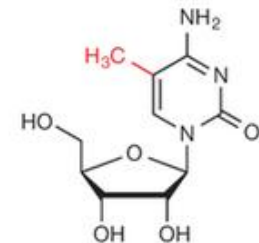
*N*⁶,2'-O-dimethyladenosine (m⁶Am)



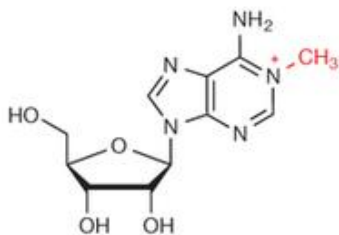
*N*⁶-methyladenosine (m⁶A)



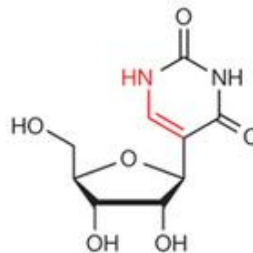
Inosine (I)



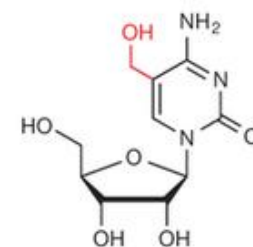
5-methylcytidine (m⁵C)



*N*¹-methyladenosine (m¹A)



Pseudouridine (Ψ)



5-hydroxymethylcytidine (hm⁵C)

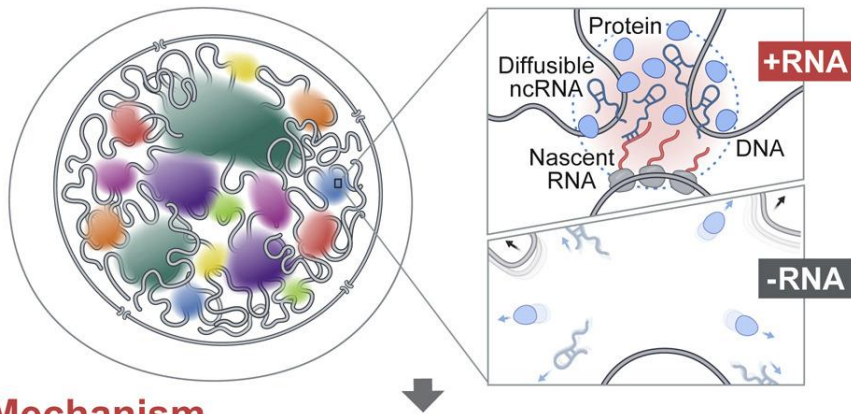
Acidi nucleici- Struttura-Interazioni-Funzioni

RNA

RNA & DNA SPRITE

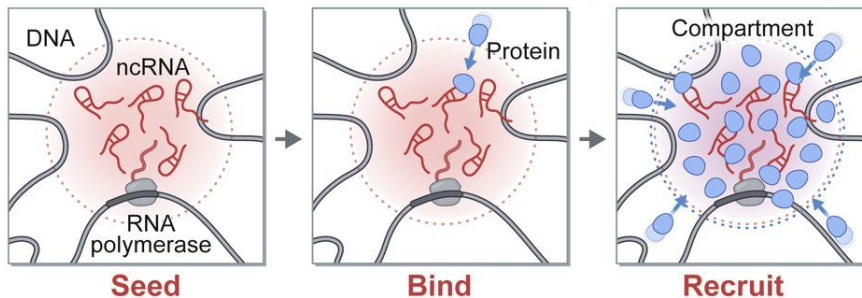
RNA demarcates spatial territories throughout the nucleus

RNA required to organize nuclear compartments

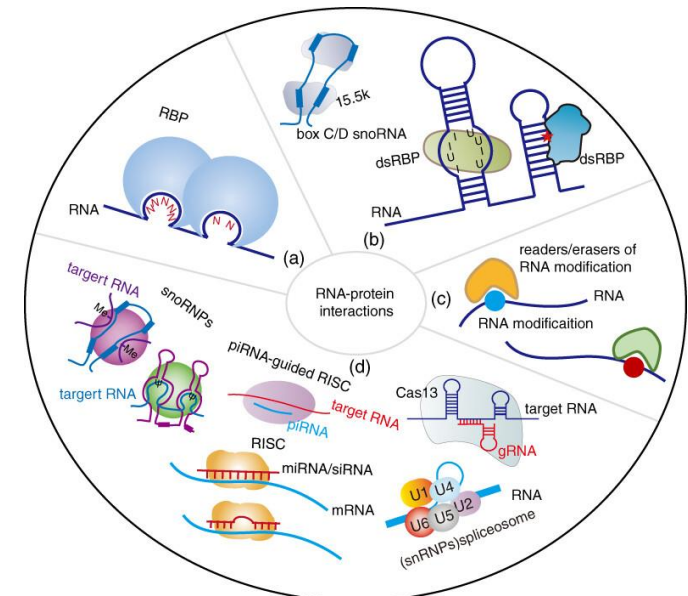
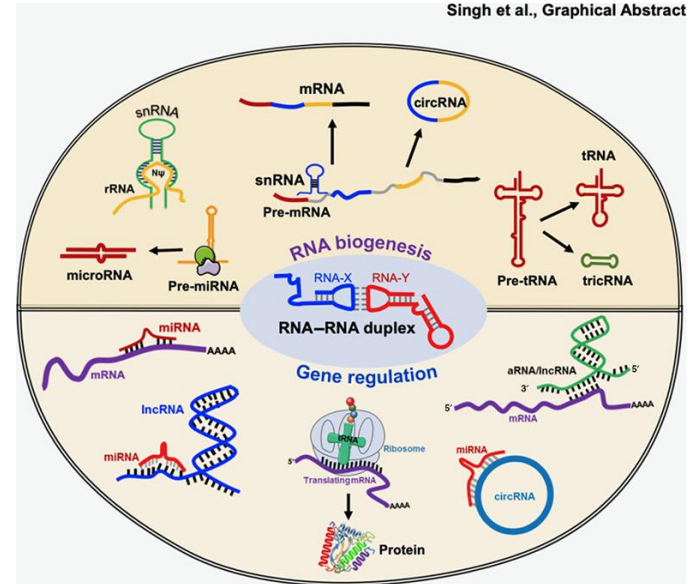


Mechanism

A shared mechanism of RNA-mediated compartment assembly



Singh et al., Graphical Abstract



Proteine

Gli aminoacidi sono i monomeri costituenti le proteine

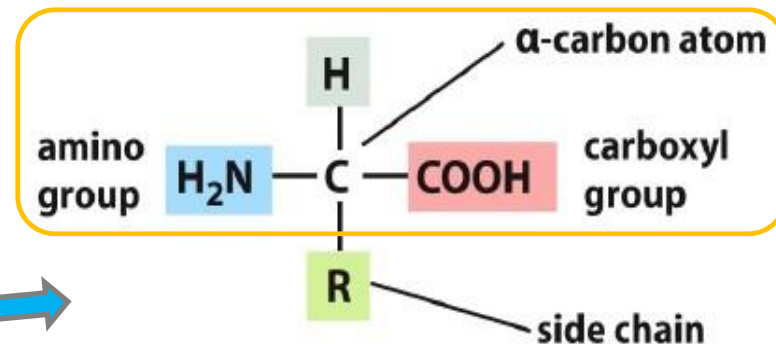
Gli aminoacidi sono 20

Sono composti da:

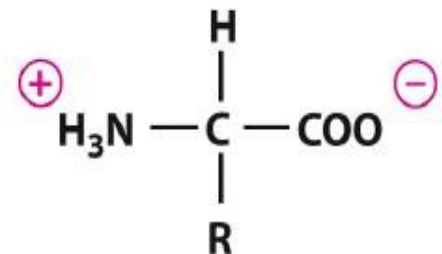
- Una struttura comune
- Gruppi R variabili

THE AMINO ACID

The general formula of an amino acid is



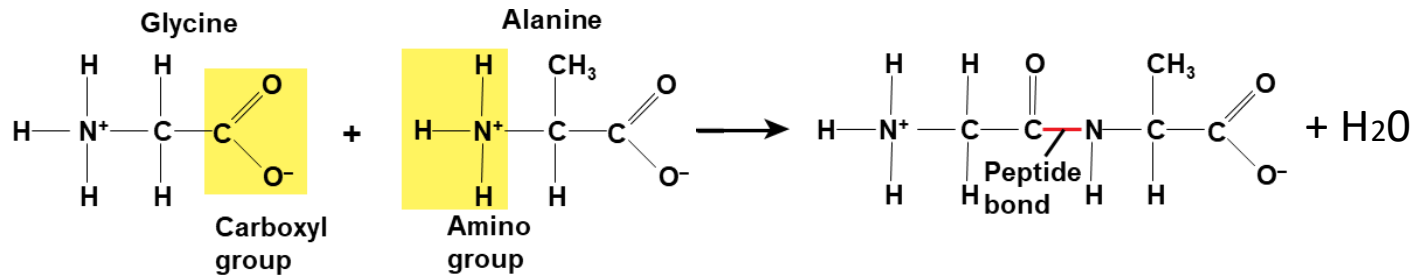
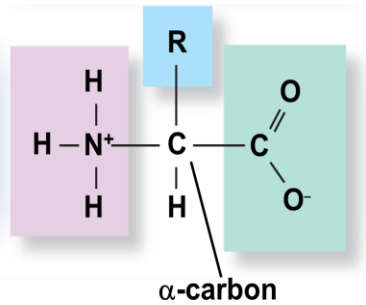
R is commonly one of 20 different side chains. At pH 7, both the amino and carboxyl groups are ionized.



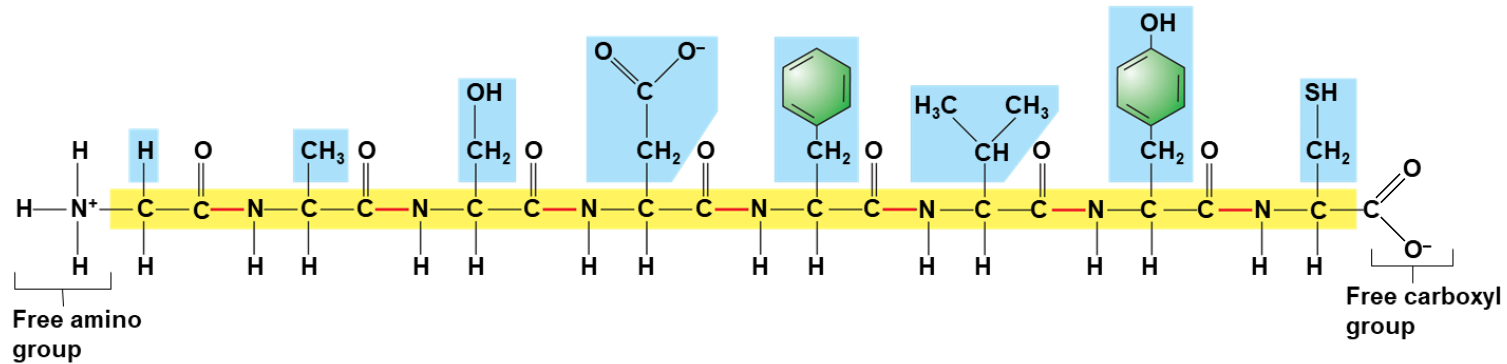
Proteine - Aminoacidi



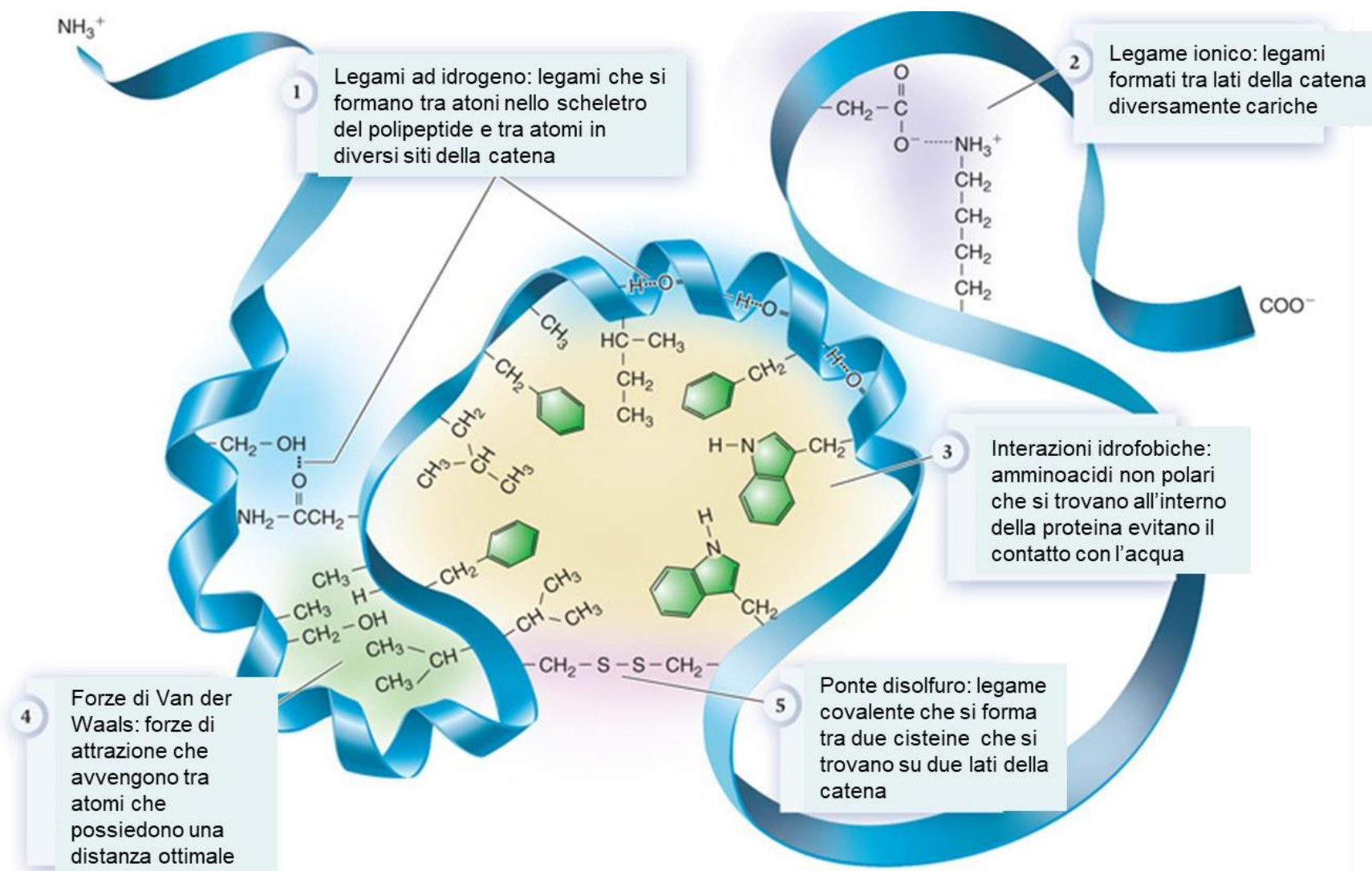
Proteine – Peptidi e Polipeptidi



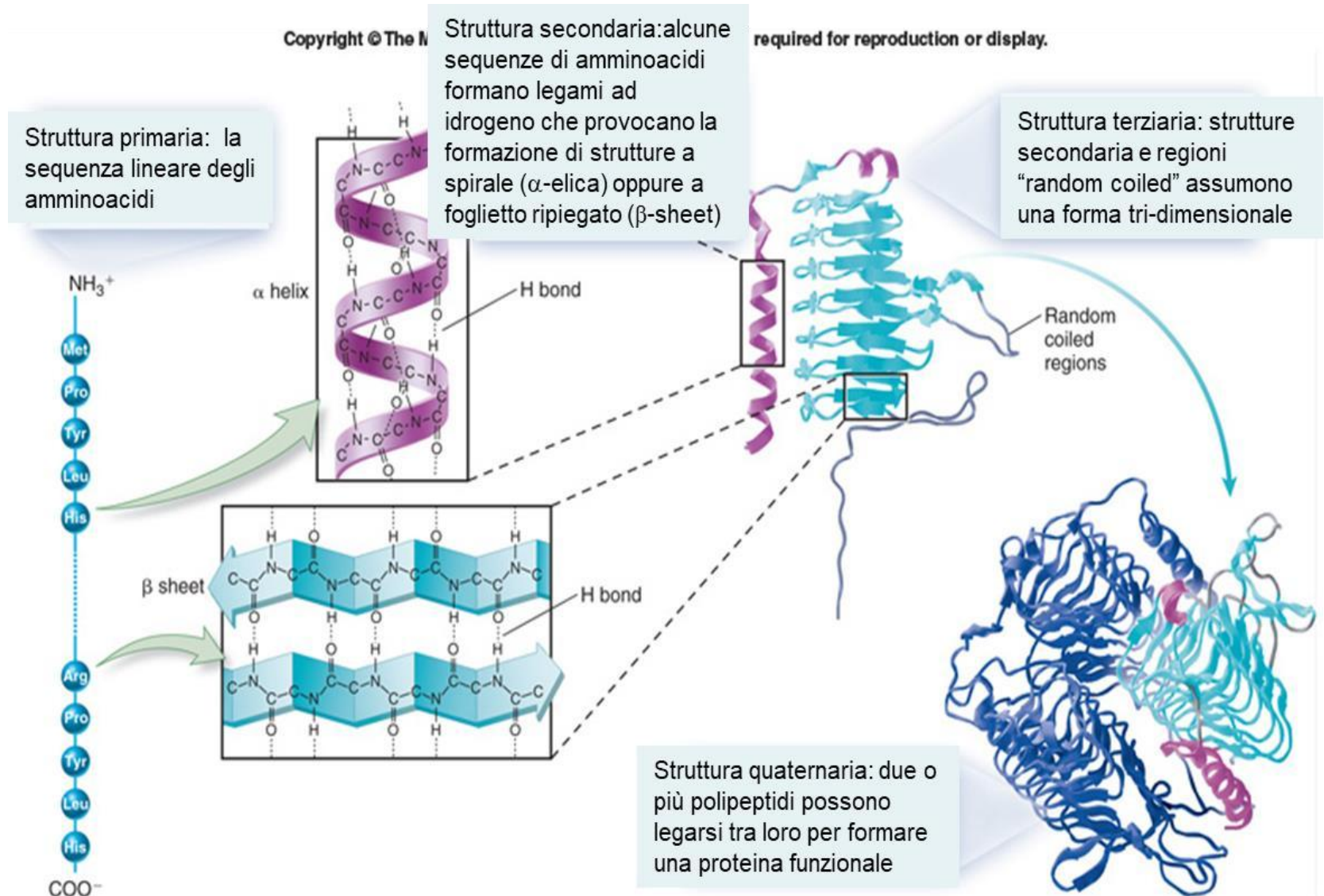
(a) Formation of a peptide bond between 2 amino acids



Un **peptide** è costituito da due o più amminoacidi legati tra loro da un legame peptidico a formare una catena composta di norma da 2 – 70 amminoacidi. Catene più lunghe di 70 amminoacidi vengono definite polipeptidi.



Proteine – Strutture dei peptide e polipeptidi

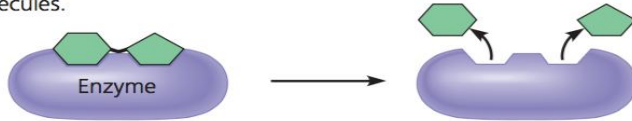


Proteine – Funzioni

Enzymatic proteins

Function: Selective acceleration of chemical reactions

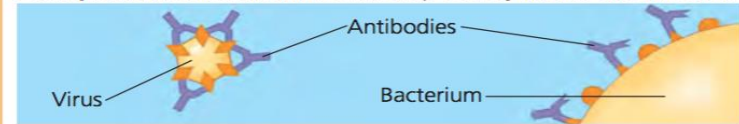
Example: Digestive enzymes catalyze the hydrolysis of bonds in food molecules.



Defensive proteins

Function: Protection against disease

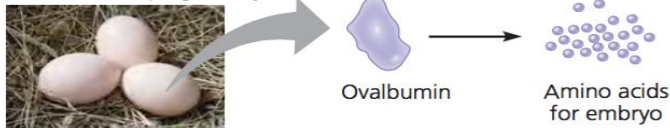
Example: Antibodies inactivate and help destroy viruses and bacteria.



Storage proteins

Function: Storage of amino acids

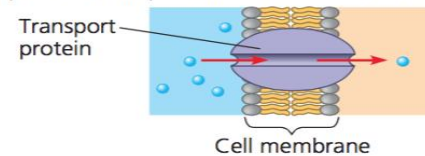
Examples: Casein, the protein of milk, is the major source of amino acids for baby mammals. Plants have storage proteins in their seeds. Ovalbumin is the protein of egg white, used as an amino acid source for the developing embryo.



Transport proteins

Function: Transport of substances

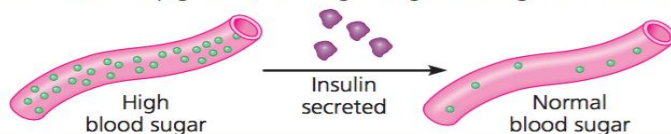
Examples: Hemoglobin, the iron-containing protein of vertebrate blood, transports oxygen from the lungs to other parts of the body. Other proteins transport molecules across cell membranes.



Hormonal proteins

Function: Coordination of an organism's activities

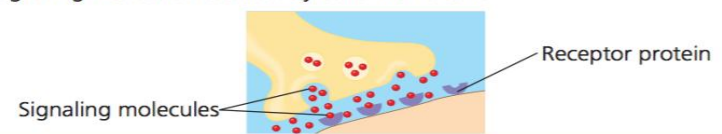
Example: Insulin, a hormone secreted by the pancreas, causes other tissues to take up glucose, thus regulating blood sugar concentration.



Receptor proteins

Function: Response of cell to chemical stimuli

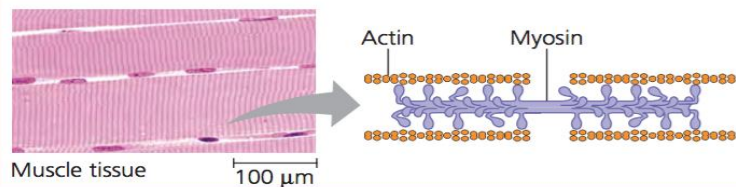
Example: Receptors built into the membrane of a nerve cell detect signaling molecules released by other nerve cells.



Contractile and motor proteins

Function: Movement

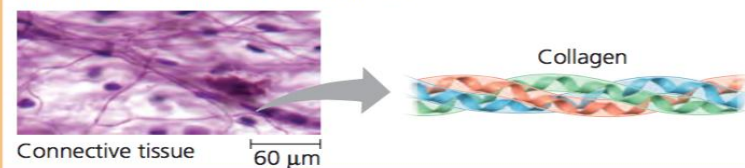
Examples: Motor proteins are responsible for the undulations of cilia and flagella. Actin and myosin proteins are responsible for the contraction of muscles.



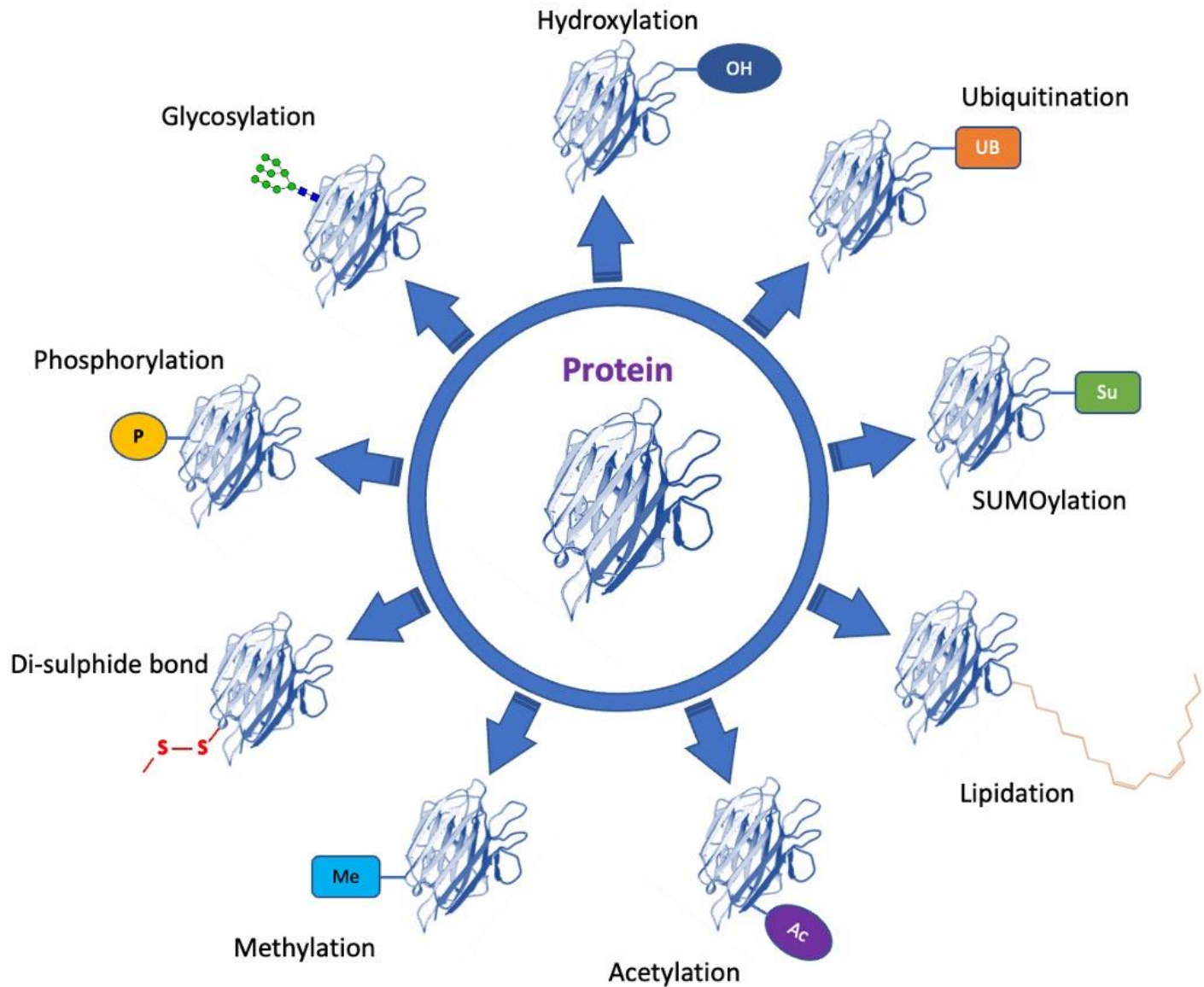
Structural proteins

Function: Support

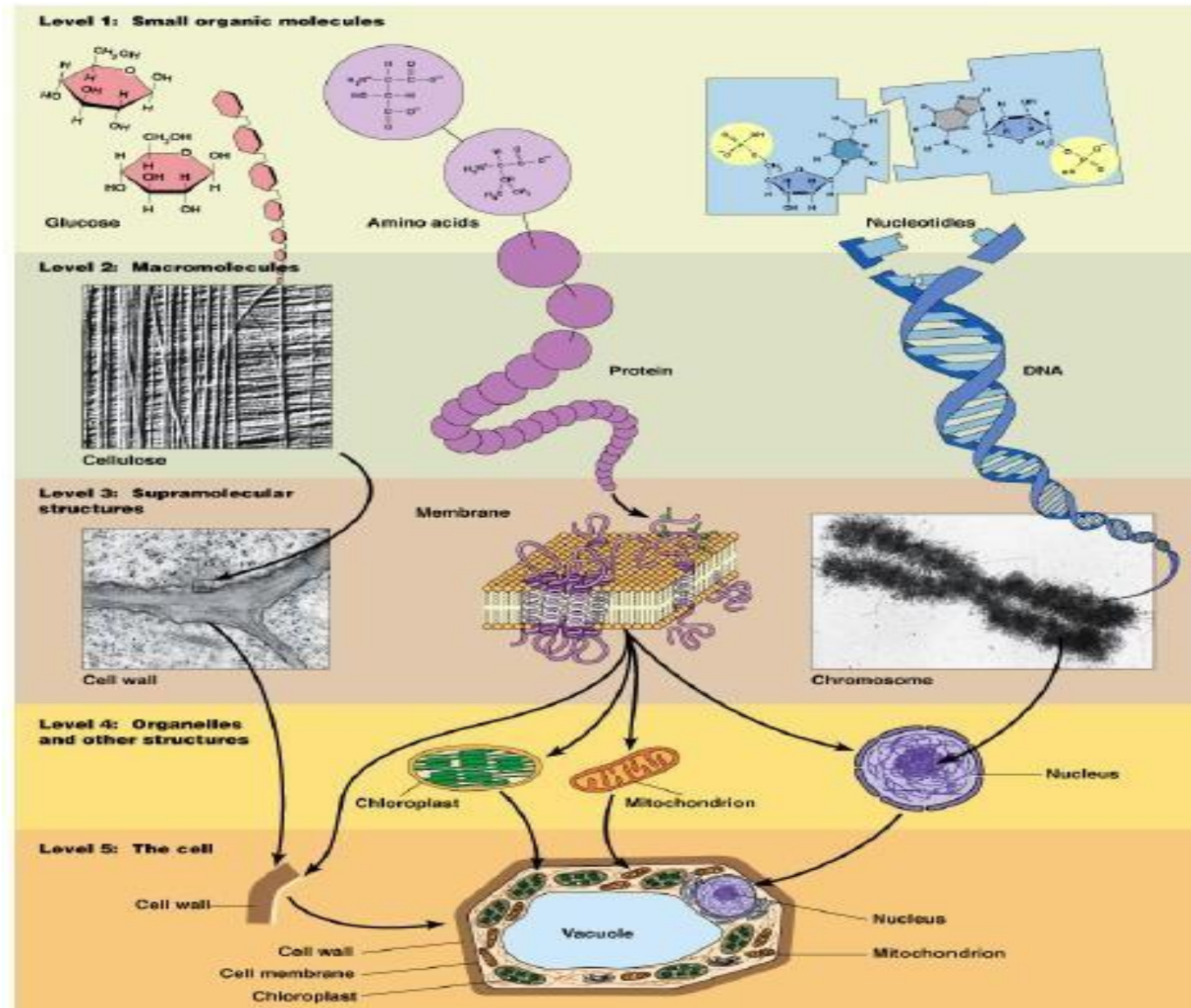
Examples: Keratin is the protein of hair, horns, feathers, and other skin appendages. Insects and spiders use silk fibers to make their cocoons and webs, respectively. Collagen and elastin proteins provide a fibrous framework in animal connective tissues.



Proteine – Modificazioni post-traduzionali

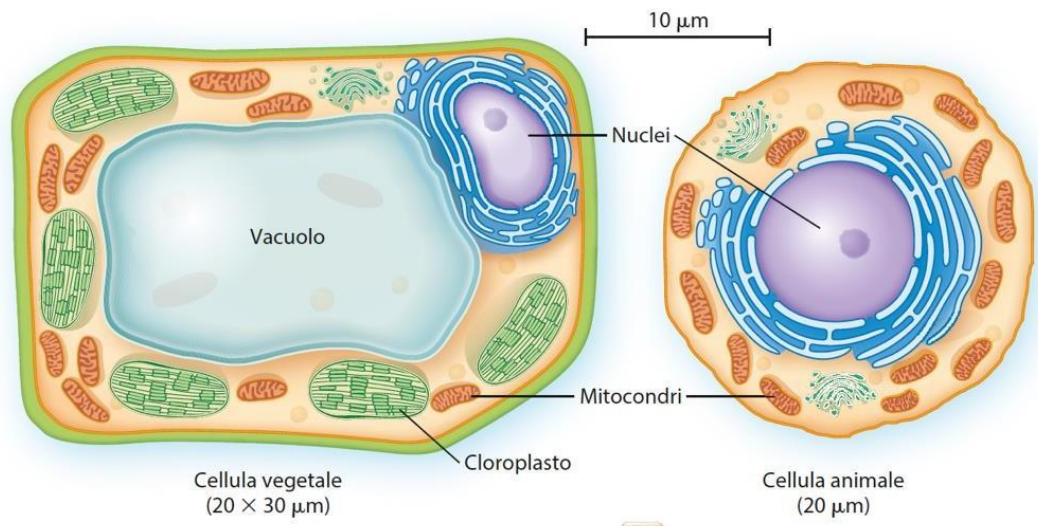


Da molecole semplici a strutture complesse

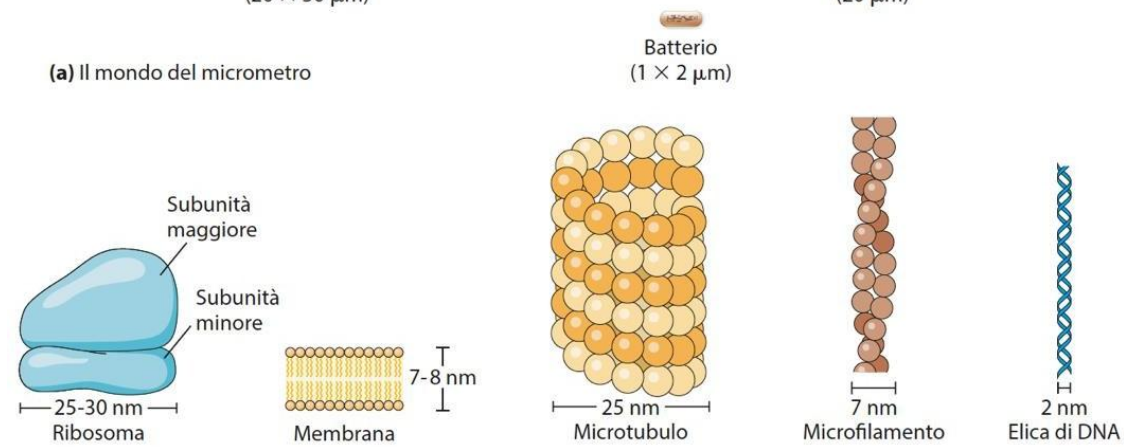


The cell: basic unit of life

Cellule, organelli e molecole: tra micrometri e nanometri



(a) Il mondo del micrometro



(b) Il mondo del nanometro

